

SunoikisisDC – 2025-2026

Digital Approaches to Cultural Heritage **3D and 2½D Imaging**

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5 February 2026

Outline

1. Multi-dimensional Imaging Techniques

- Which 3D approach?
- Other 3D options
- 3D and 2.5D in focus

2. Research Insights from Digital Imaging

- Working with outputs
- Case Studies

3. Beyond Imaging

- Ethical considerations
- What do you need the output for? (Discussion)

4. Exercise presentation

Multi-dimensional Imaging Techniques

Which 3D approach?

Other 3D options

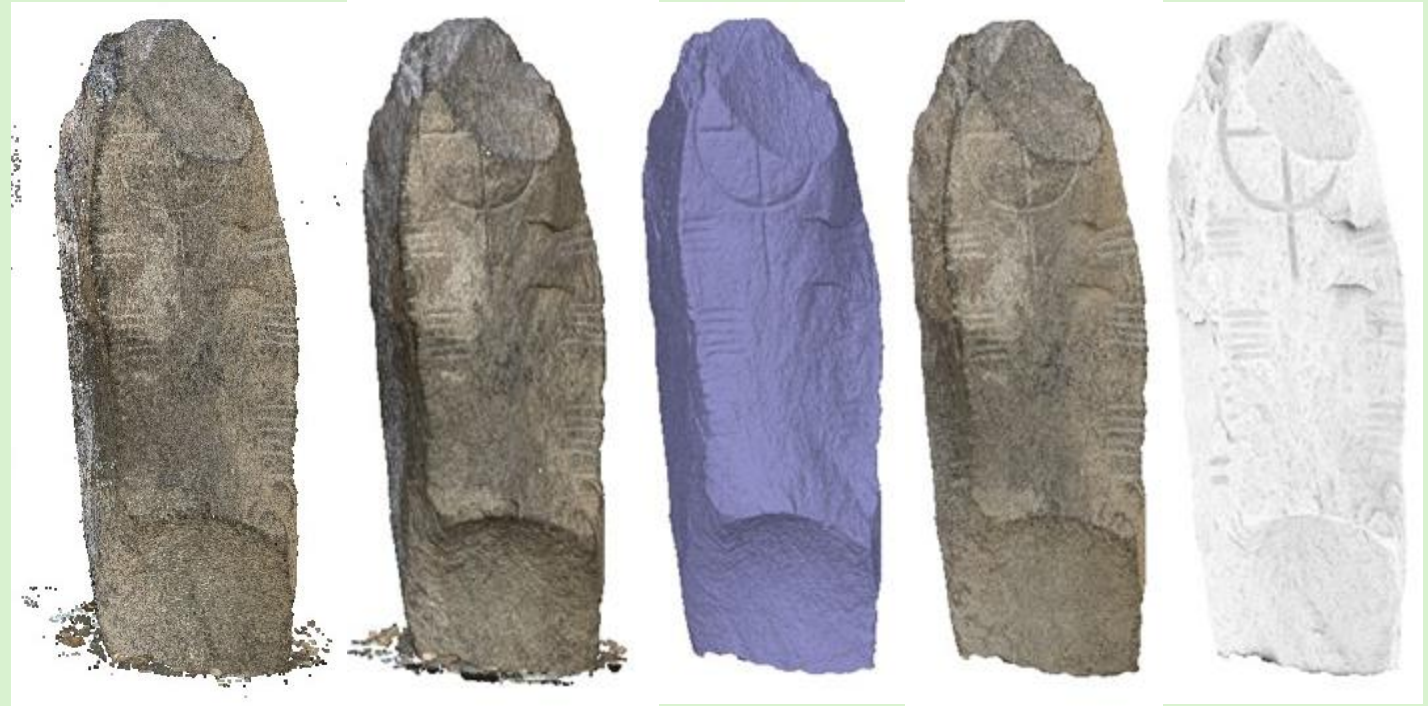
3D and 2.5D in focus

Multi-dimensional Imaging Techniques

Which 3D approach?

3D Data Capture

- Many methods
 - Laser scanning
 - Structured Light Scanning
 - Artec vs Zeiss
 - CT Scanning
 - Photogrammetry
- Point clouds from measured points in 3D space
- Connect points with triangles or rectangles = mesh



Which 3D approach?

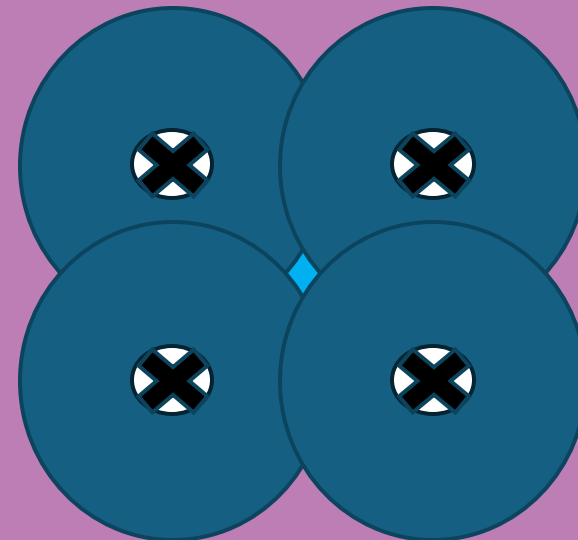
- Material of Target
 - Stone, clay, metal, organic, mixed
 - Reflectivity, translucency, surface contrast
- Scale
 - Landscape, building, monument
 - Object, fragment, inscription
- Resources: Budget & Time
 - Purchase equipment?
 - Rent equipment (day rate)?
 - Outsource
- Working Area/Environment
 - Studio vs gallery vs field
 - Access, handling limits, stability
- Purpose of 3D asset
 - High resolution mesh? Accuracy
 - Convincing texture? Legibility/Visual communication
 - Interactive? Engagement

Multi-dimensional Imaging Techniques

Other 3D Techniques

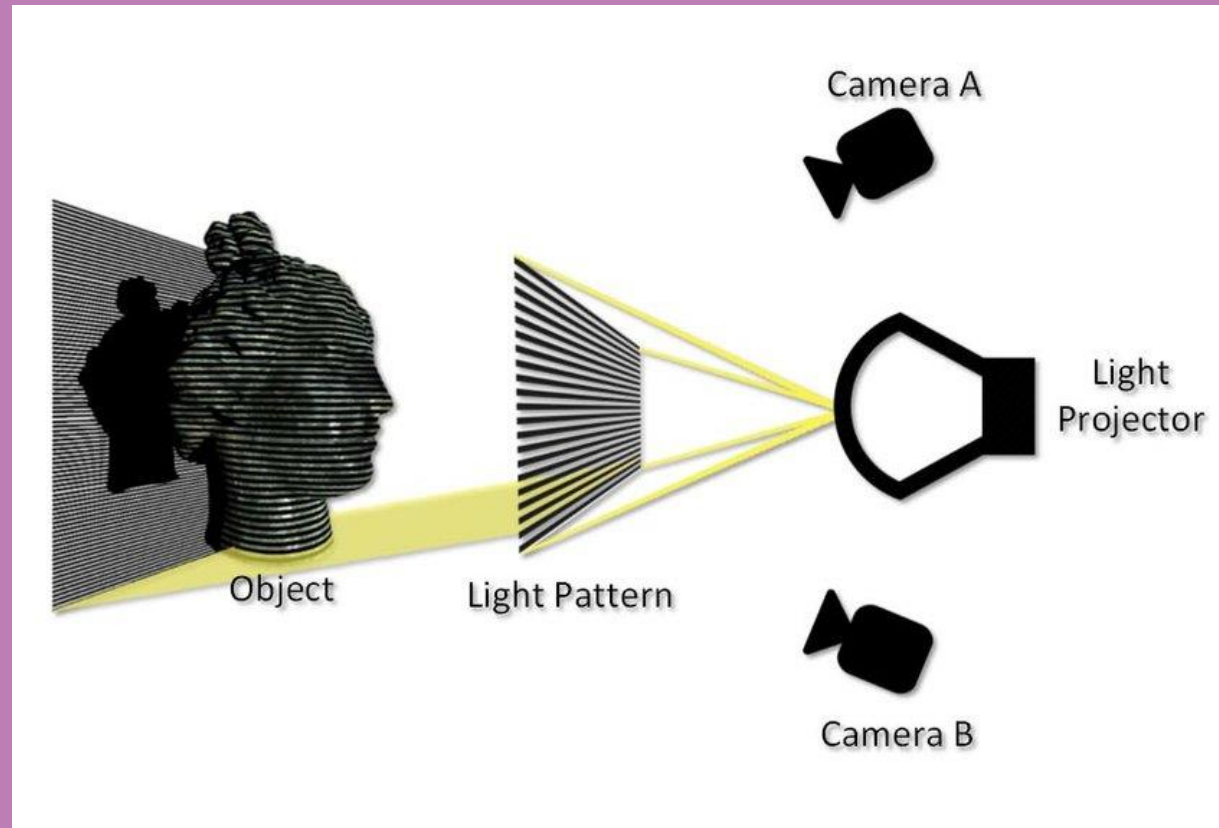
Laser scanning: ToF

- “Time of Flight”
- Passes a line of lasers over an object
- Lasers reflect from the surface
- Sensor in the scanner records how long to return
- Placed in strategic positions around the object/room to ensure coverage
- Data gap directly under scanner



White-light scanning (Structured-light)

- Handheld unit projects patterned light onto the object
- Two cameras in fixed positions measure the deviations in the light's pattern to build 3D mesh.
- Triangulation laser scanners: similar principle, but measures deformation of line of lasers
- MakerBot Digitiser
- Faro Focus3D



https://www.researchgate.net/figure/Working-principle-of-structured-light-scanner_fig2_320657865



Artec Eva

Artec



- Requires laptop to capture data
- Specific distance away



Photo: Dr Nora White

Artec Leo

- Doesn't need to be attached to a laptop
- Screen built into handheld scanner – tracks progress
- Laptop processing
- Projected light = reflection issues
- Software – updates old projects
- But limited by original data capture settings

Other types of Structured Light

- Blue
- Zeiss Comet L3D
 - Interchangeable lenses based on size of object
 - Lasers for positioning, not for data capture
 - Attached, automatic turntable
 - Permissions from Museums





Computerized Tomography (CT) Scanning

- Series of X-ray 2D photographs of different sections of the object, superimposed into layers to form a 3D model



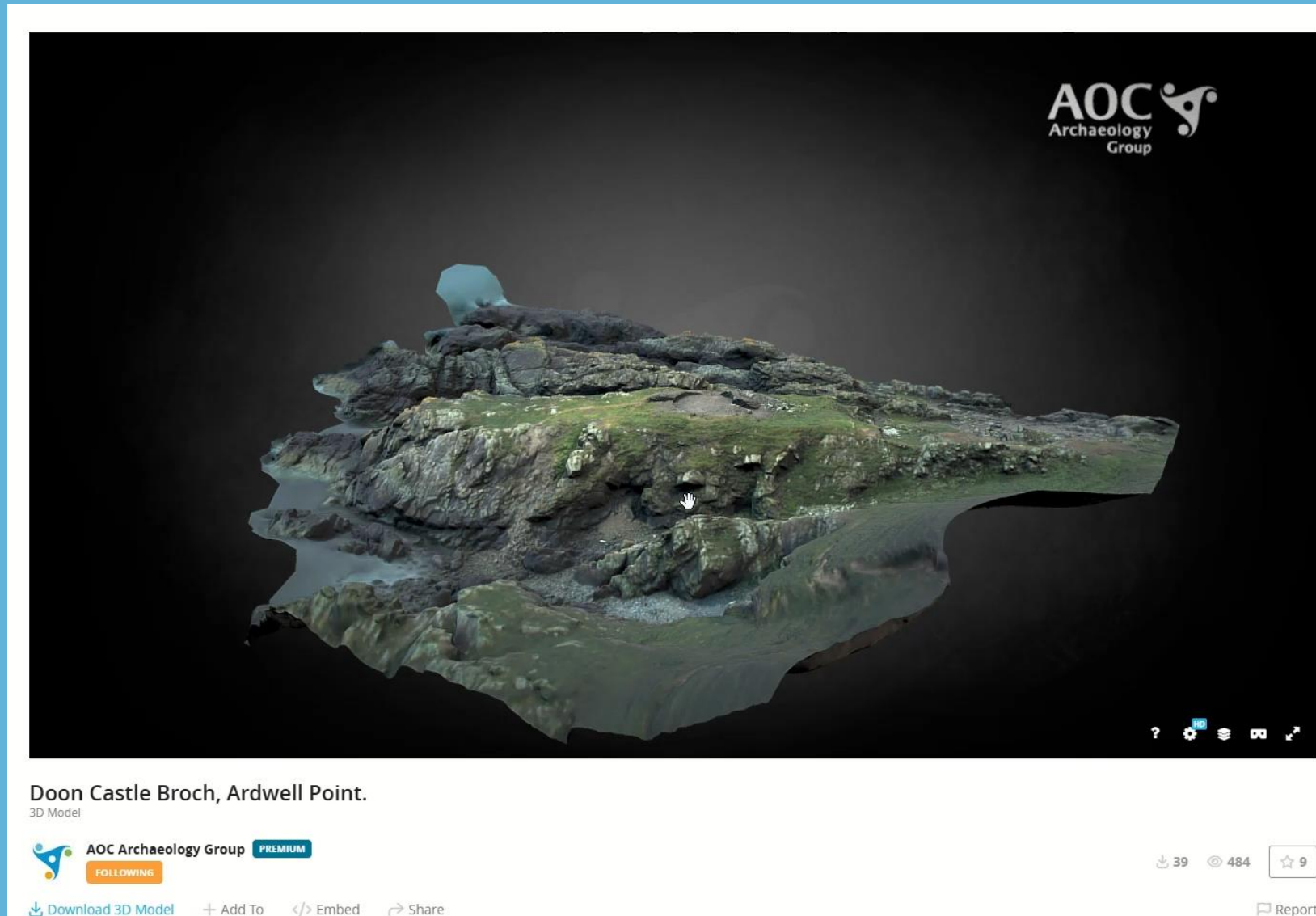
- Allows scanning of the internal geometry and structure of an object, not only the external part

Multi-dimensional Imaging Techniques

3D and 2.5D in Focus

Photogrammetry

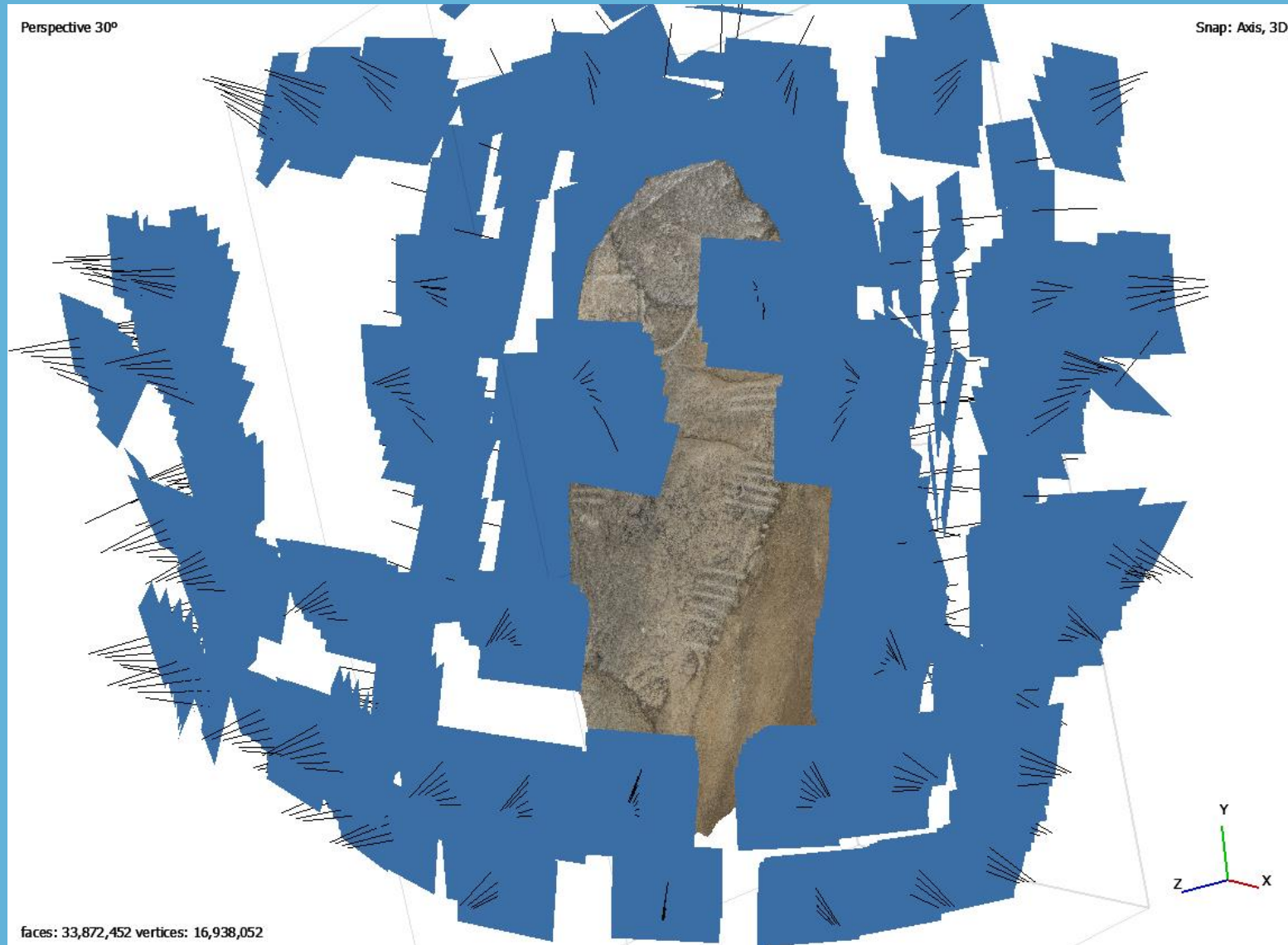
- Aerial (Drone) Photogrammetry
- **Object-focused photogrammetry**
- Phone photogrammetry



<https://skfb.ly/ozJU8>

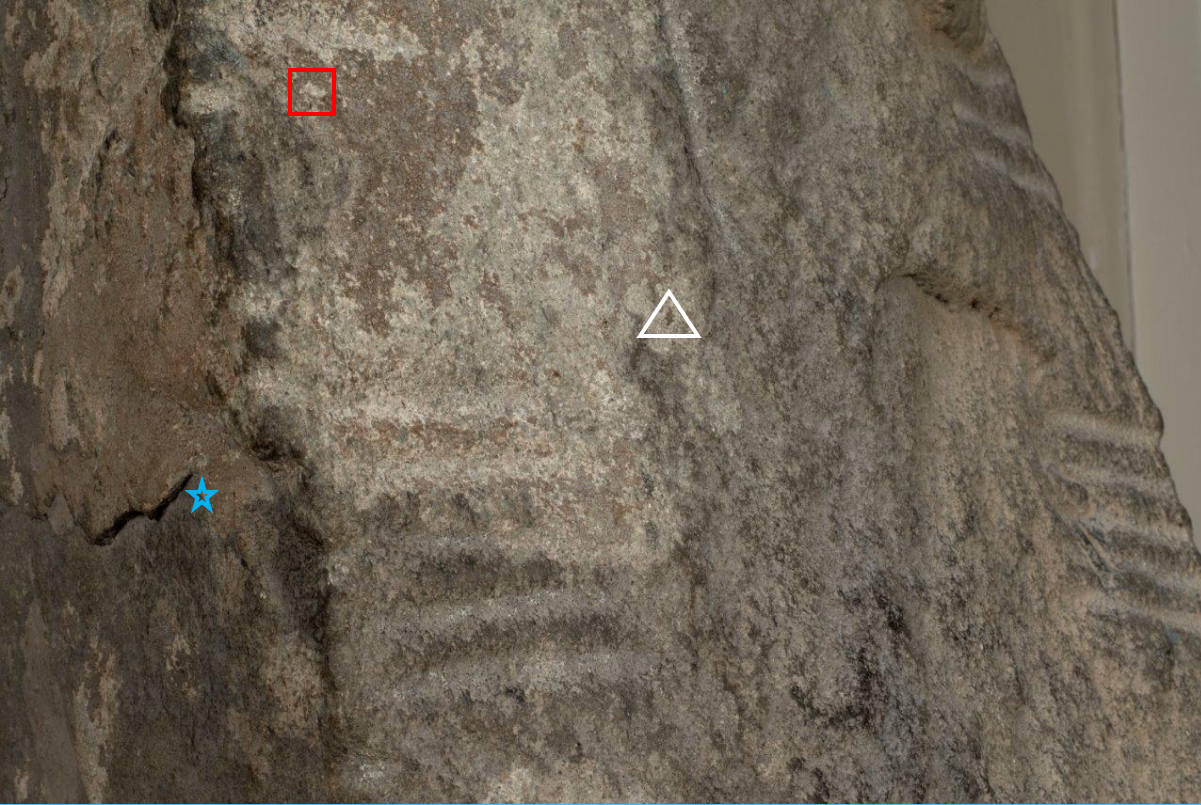
Photogrammetry

- Structure from Motion
 - Move camera around object, take photographs
- Flexible technique
 - Wide range of material to capture
 - Outdoors, indoors
- Software: Agisoft Metashape, or RealityCapture/RealityScan, or AliceVision (Open source)



I-ARM-001 Drumconwell

Photogrammetry



- At least 50% overlap between photographs
- Clarity of photographs is key
- Software finds points in common, calculates surface geometry
- 2 scale bars in several photos
 - User identification of points



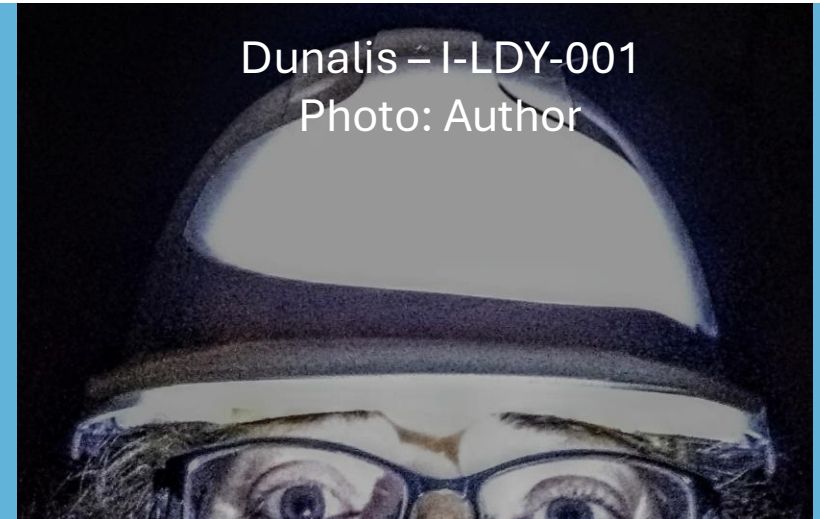
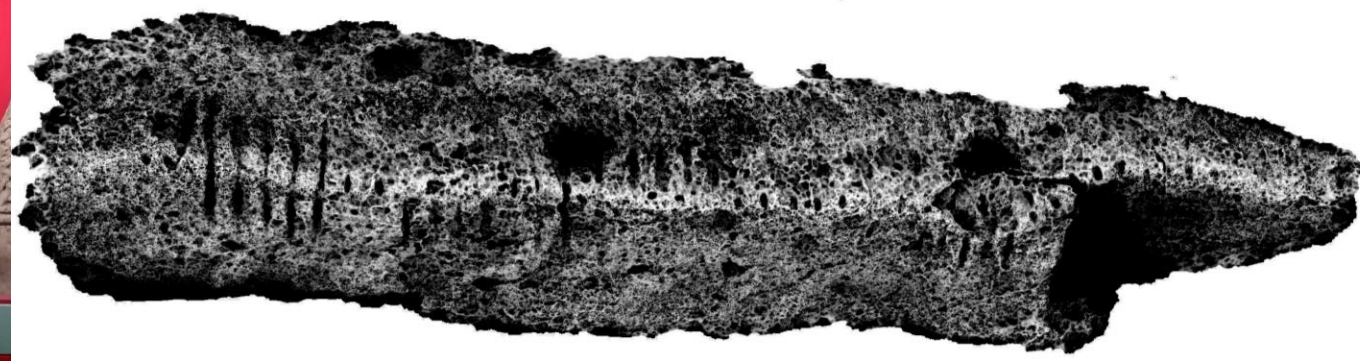
Flexibility



Altyre Stone – S-MOR-002
Photo: Dr Nora White



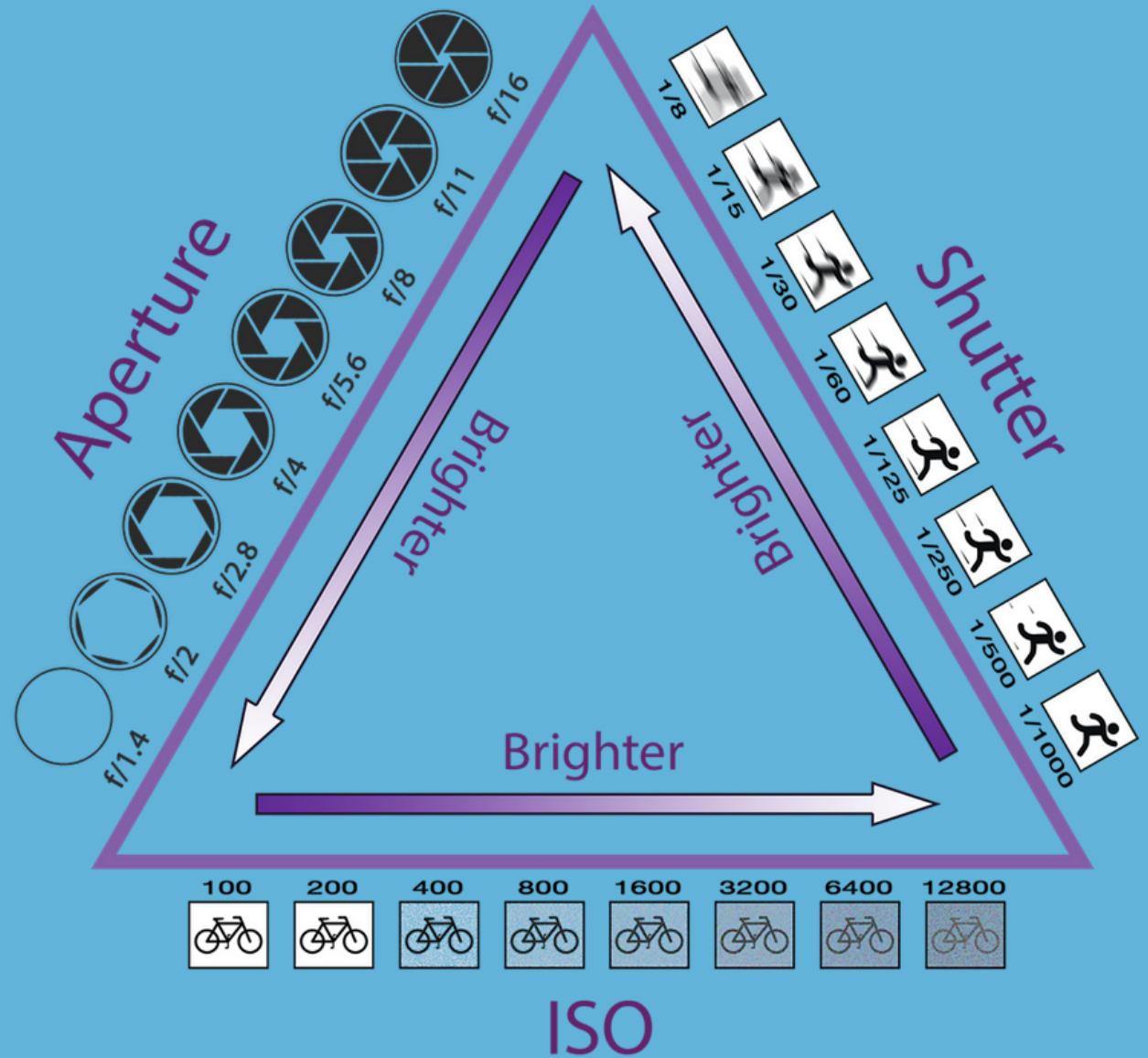
Formaston – S-ABD-003
Photo: Author



Dunalis – I-LDY-001
Photo: Author

Better Photography, better 3D model

- Good depth of field, rule of thumb:
 - Low ISO, High F-Stop/Aperture, variable Shutter Speed
 - Software needs to 'see' surface to make connections



<https://blog.garrethughes.com/understanding-the-exposure-triangle-iso-aperture-shutter-speed-75db21958cc7>



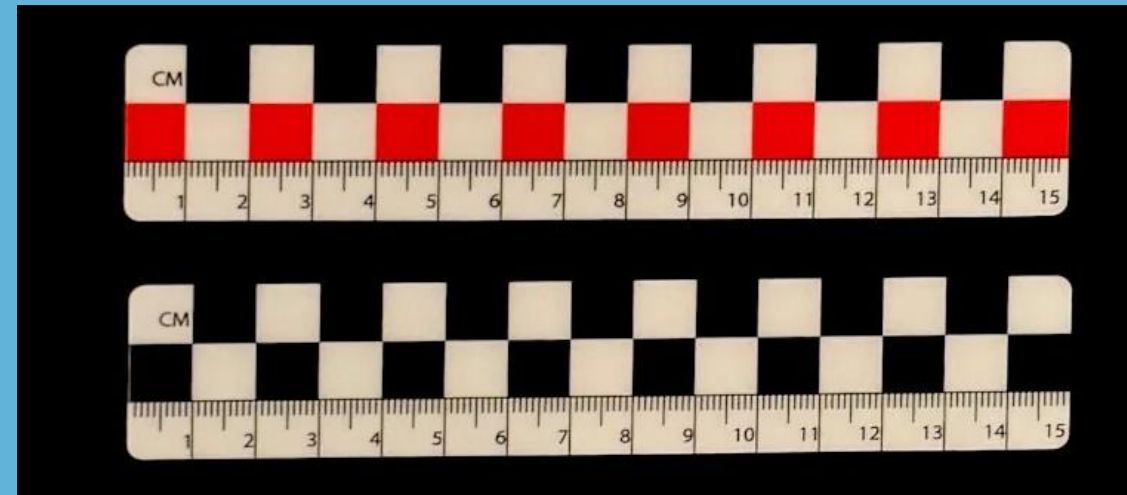
Aperture f/1.4 Using 50mm lens and full-frame DSLR
Alex1ruff, CC BY-SA 4.0
<<https://creativecommons.org/licenses/by-sa/4.0>>, via
Wikimedia Commons



Aperture f/22 Using 50mm lens and full-frame DSLR
Alex1ruff, CC BY-SA 4.0
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Wikimedia Commons

Photogrammetry Equipment

- DSLR Camera (more MP, fewer photos you need to take, in theory)
 - Shoot in RAW, but if you have laptop to test photos, jpeg-fine as well
- Tripod
- Remote trigger (app or physical, or both)
- Two Lights (Indoor photogrammetry)
- Contrast background
- Turntable
- Two flat scale bars with easy-to-identify corners (not identical)
- Blue-tac (not for artefact! Ensures scale bars stay in same place)

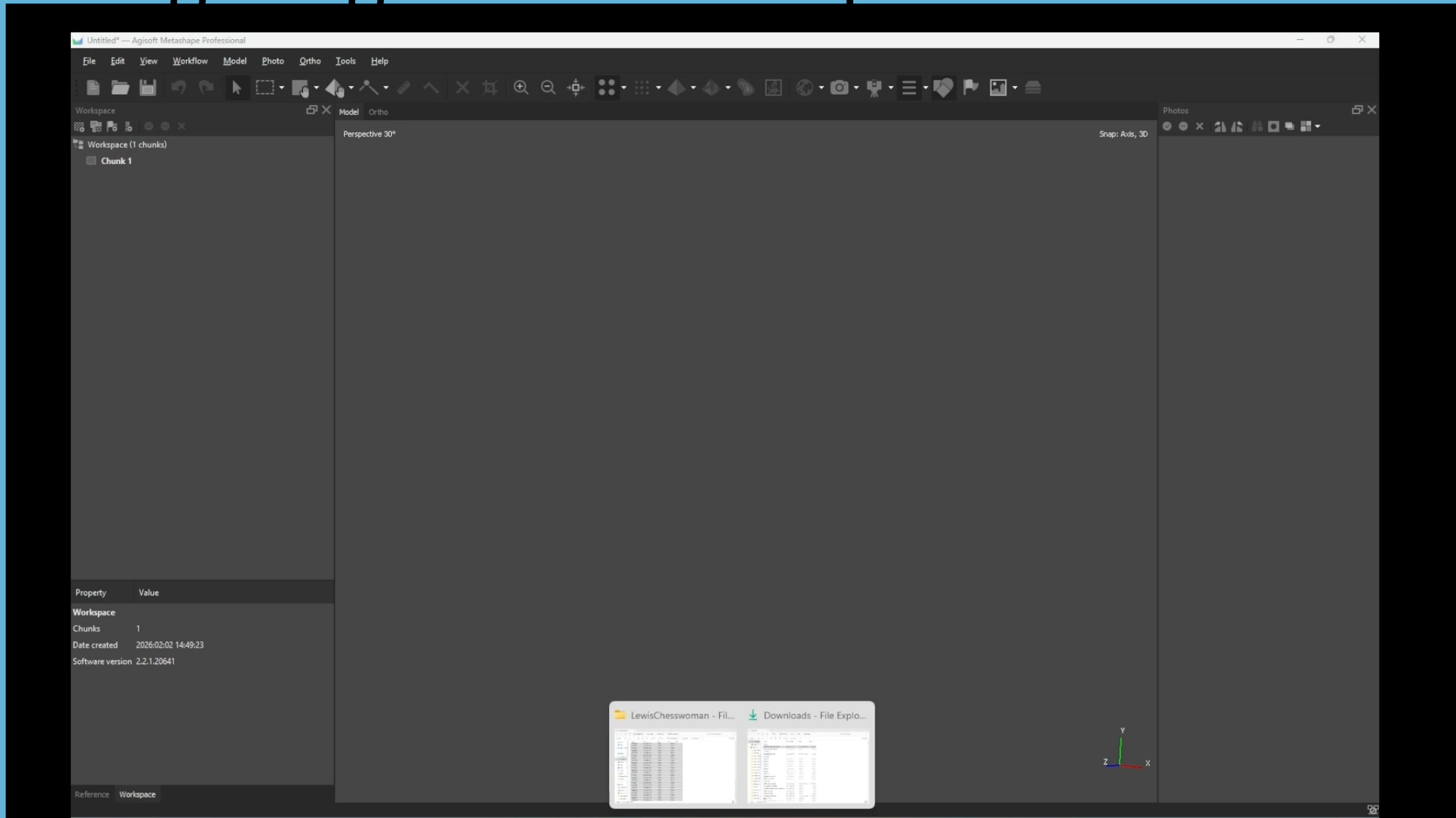


Photogrammetry Data Capture Workflow



Unlisted video, link: <https://youtu.be/qDf8OPXzCWA>

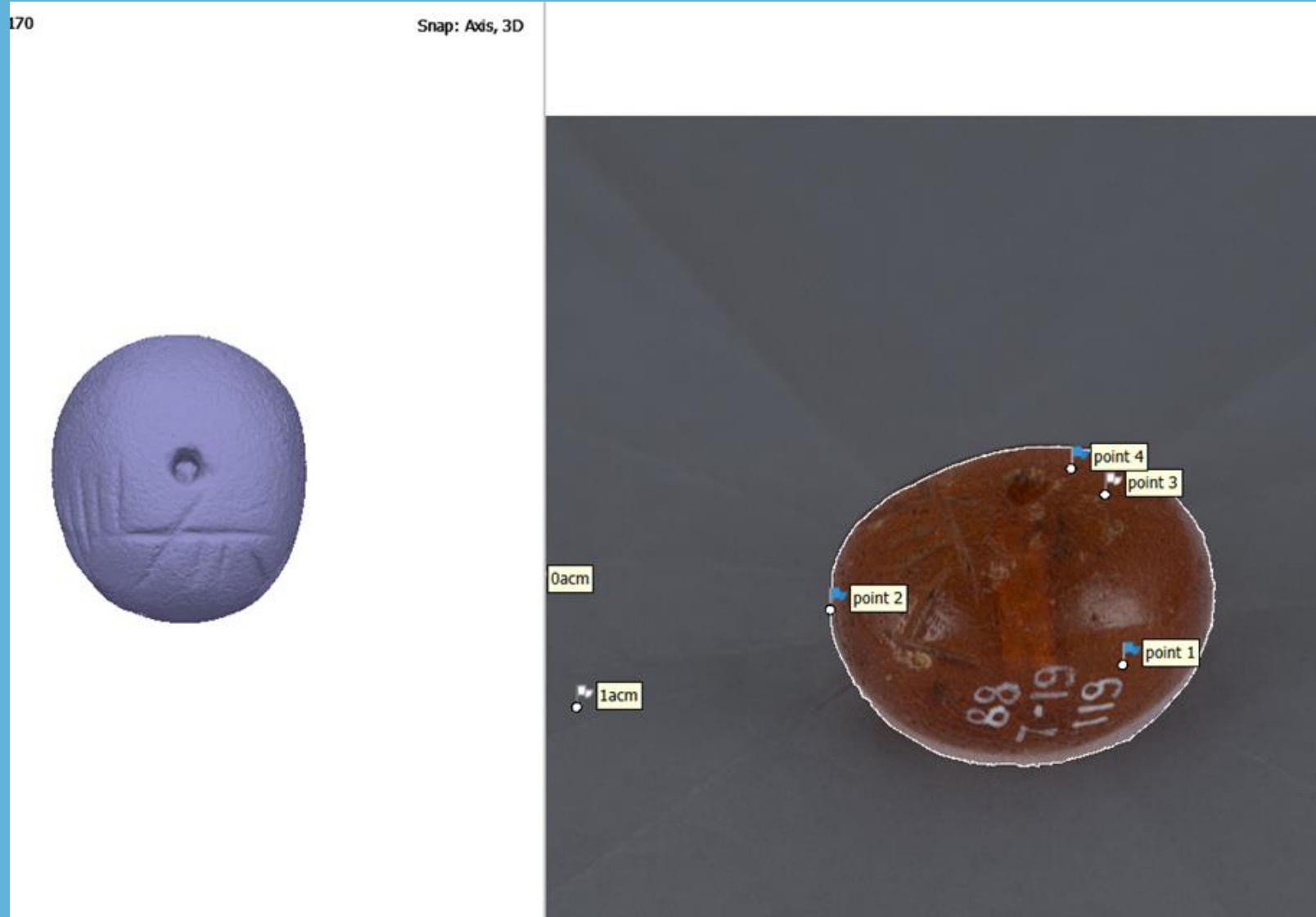
Processing in Agisoft Metashape



- <https://dinosaurpalaeo.wordpress.com/2015/10/11/photogrammetry-tutorial-11-how-to-handle-a-project-in-agisoft-photoscan/>
- Unlisted video, link: <https://youtu.be/eUQltvrbHkg>

Masking

- Software considers background too
 - Issue for turntables with background visible, interferes with alignment
 - 'Block' with masking



Photogrammetry Phone Apps

- Outreach activities
 - Less invasive than transferring images from phones
- Subscription or free?
 - Changes over time - Polycam
- Cloud processing?
 - Automatic?
- Display website?
- Archive/Metadata/Paradata?
- Polycam: free for 7 days, subscription after
- KiriEngine: Free, only pay for advanced post-capture options

Software

- Using Outputs

- Meshlab: <https://www.meshlab.net/>

- Open Source
 - Processing and editing 3D mesh
 - Tutorials available on YT, necessary as quite complicated
 - Better measuring tool

- Blender: <https://www.blender.org/>

- Open Source
 - Animations, 2D scenes, 3D scenes, texturing, etc.
 - Used widely, tutorials available
 - Higher spec computer recommended

- Autodesk

- 3DS Max

- Professional 3D editing, rendering and animation software
 - £££

- Zbrush

- Digital sculpting software

- ££

- Adobe Photoshop/Camera RAW

- Batch white balancing of photos

- Subscription: ££

Compare the 3D Results!

- British Museum Mummy Mask (8 years ago):
 - Photogrammetry: <https://skfb.ly/6s6xU>
 - Artec Spider (Structured-Light): <https://skfb.ly/onBqS>
 - CT Scan: <https://skfb.ly/69pyS>
- More combinations of methods as technology develops
 - Understand how underlying methods take measurements



2.5D Approaches

Why “2.5D”?

- Focus on surface behaviour, not full volume
- Surface information derived from light–surface interaction
- Enhanced relief and legibility without full 3D geometry
- Optimised for inscriptions, textured surfaces, worn or low-contrast surfaces

Photometric Stereo Imaging (PSI)

- Dense, per-pixel surface normals
- High-resolution, controlled capture

Reflectance Transformation Imaging (RTI)

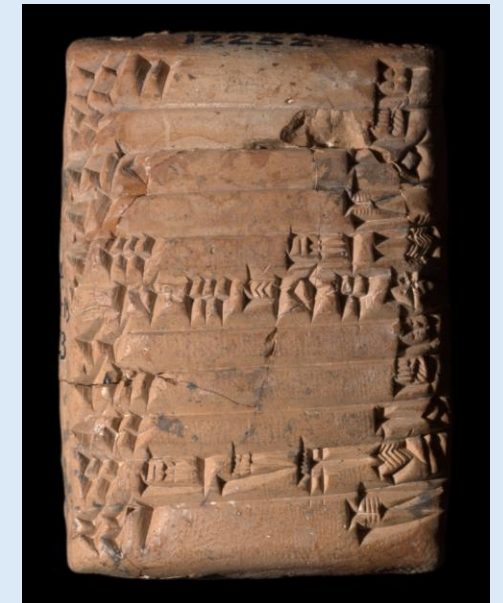
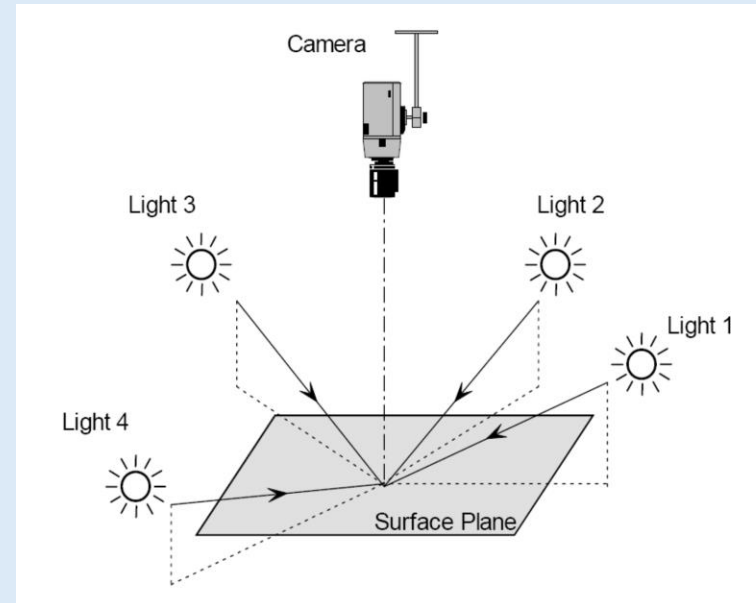
- Interactive relighting from photographic sets
- Flexible, low-overhead deployment

Photometric Stereo Imaging (PSI): Principles

- Fixed camera, fixed object
- Multiple images under controlled, changing light directions
- Computes per-pixel surface orientation (surface normals)
- Focuses on how light interacts with surface micro-topography
- PSI $\neq$ photography
- PSI $\neq$ 3D scanning
- PSI = computational interpretation of light behaviour



Inscribed tablet (BM 12252) ©Trustees of the British Museum



<https://www.sci.utah.edu/~gerig/CS6320-S2013/Materials/CS6320-CV-S2013-Photometric-Stereo-updated.pdf>

Photometric Stereo Imaging (PSI): Capture

Fixed setup

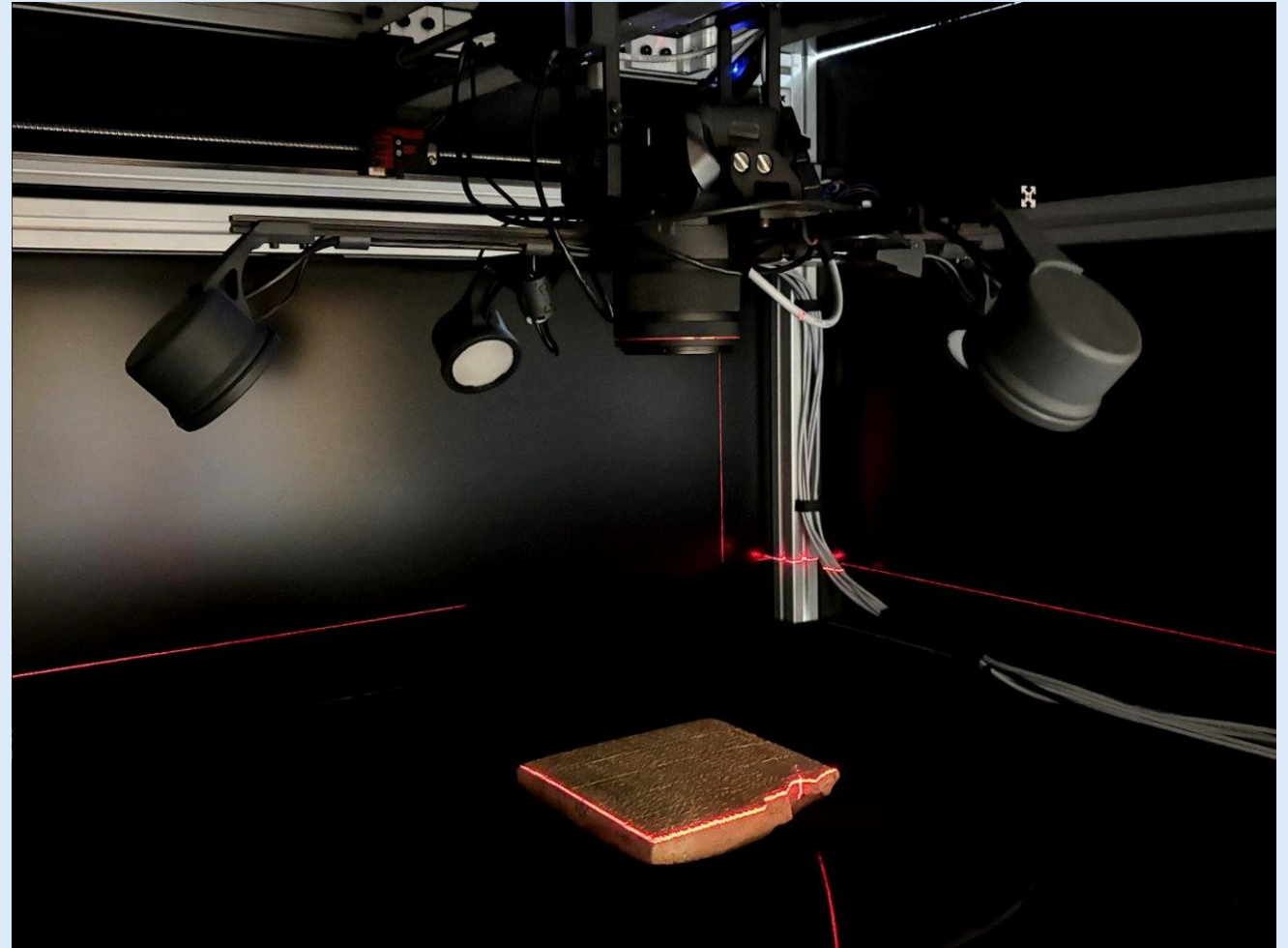
- Fixed camera (single viewpoint) and target
- Camera settings unchanged throughout capture
- Manual exposure, fixed white balance

Controlled lighting

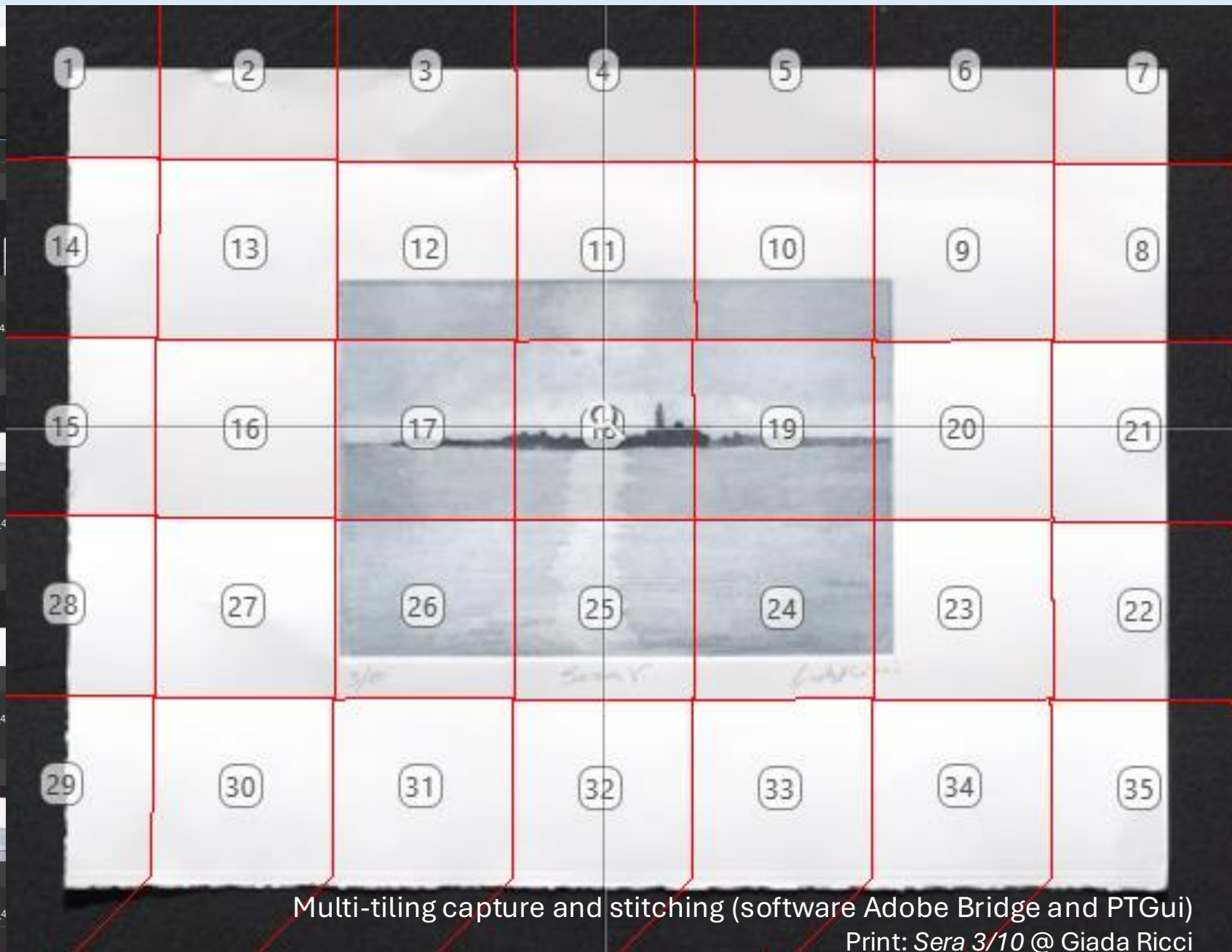
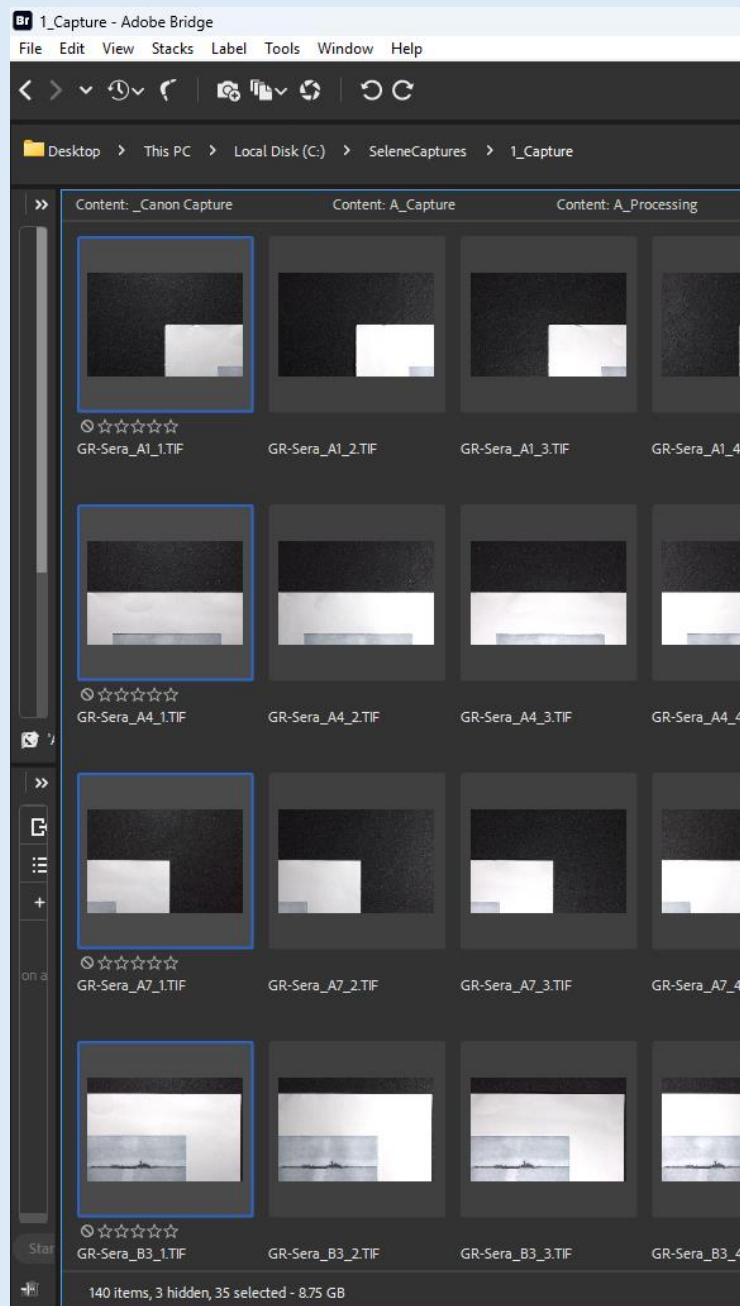
- Light direction changes between shots (one light at a time or controlled LED array)
- Light positions are known or calibrated

Image acquisition

- Multiple images of the same surface
- Identical framing across all frames



Photometric Stereo Capture using the Selene PSS
Mathilde Touillon-Ricci / ©Trustees of the British Museum

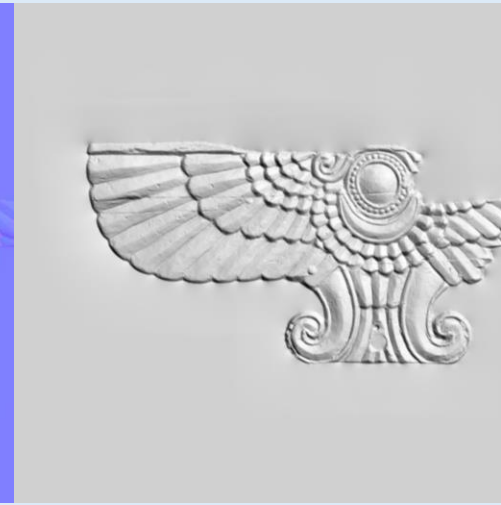
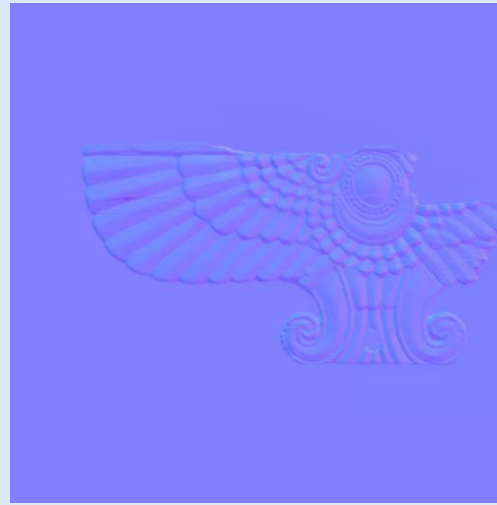


Multi-tiling capture and stitching (software Adobe Bridge and PTGui)

Print: Sera 3/10 @ Giada Ricci

Photometric Stereo Imaging (PSI): Outputs

- Surface normal maps
(orientation field = orientation at each pixel)
- Albedo (computational reflectance estimate = how bright the target material is)
- Depth-like visualisations
(derivatives $\neq$ mesh)
- Enhanced renders
(shading that enhances legibility > layered composite)
- Outputs are interpretative surfaces, not objects



Top left: normal map
Top centre: depth map
Top right: albedo

Bottom:
layered composite
(albedo + depthmap)

Ivory fitting (BM 127176) ©Trustees of the British Museum

Photometric Stereo Imaging (PSI): Summary

PSI is designed for ultra-high-resolution capture of surface behaviour on objects that are flat or semi-flat

- Shallow relief legibility: inscriptions, tool marks, wear, erasures, rewritings
- Non-contact, highly controlled capture
- Research questions consider how something was written or made, object surface information

Strengths

- Shadows and highlights become data
- Separates colour (albedo) from surface information (depth map)
- Produces multiple complementary outputs from one capture

Limits

- Relief is deep or strongly undercut, recesses dominate the object
- Highly reflective or metallic surfaces (unless polarised)

From PSI to RTI:

- PSI and RTI sit on the same methodological level: they both use multi-light capture to support computational relighting
- They differ in:
 - capture control
 - computational modelling
 - types of outputs

Multi-dimensional Imaging Techniques

- 3D and 2.5D in Focus - RTI

RTI

- Finely carved artefacts – inscription not always clear in 3D model
- Camera in one position
 - Same settings
 - Different lighting angle
 - Sphere tracks light position
 - Software calculates how each pixel reacts to different light positions

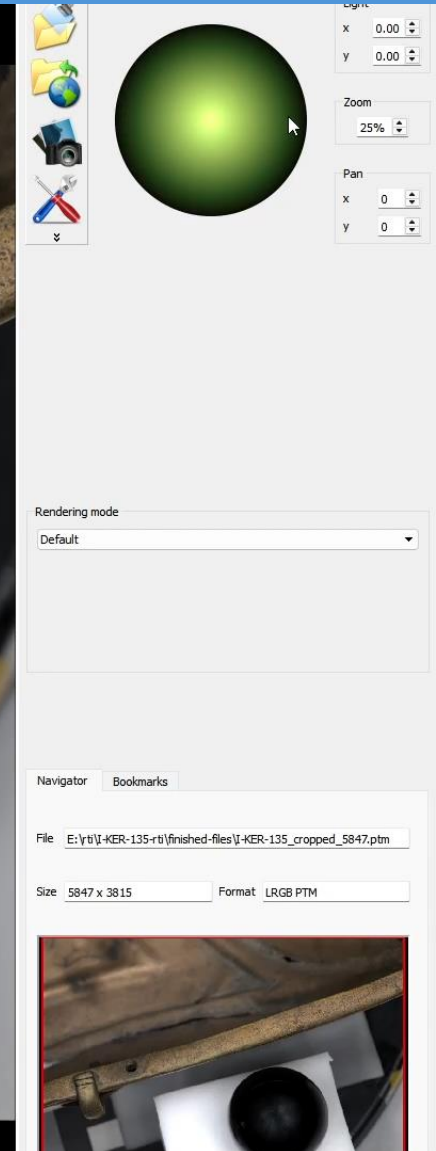
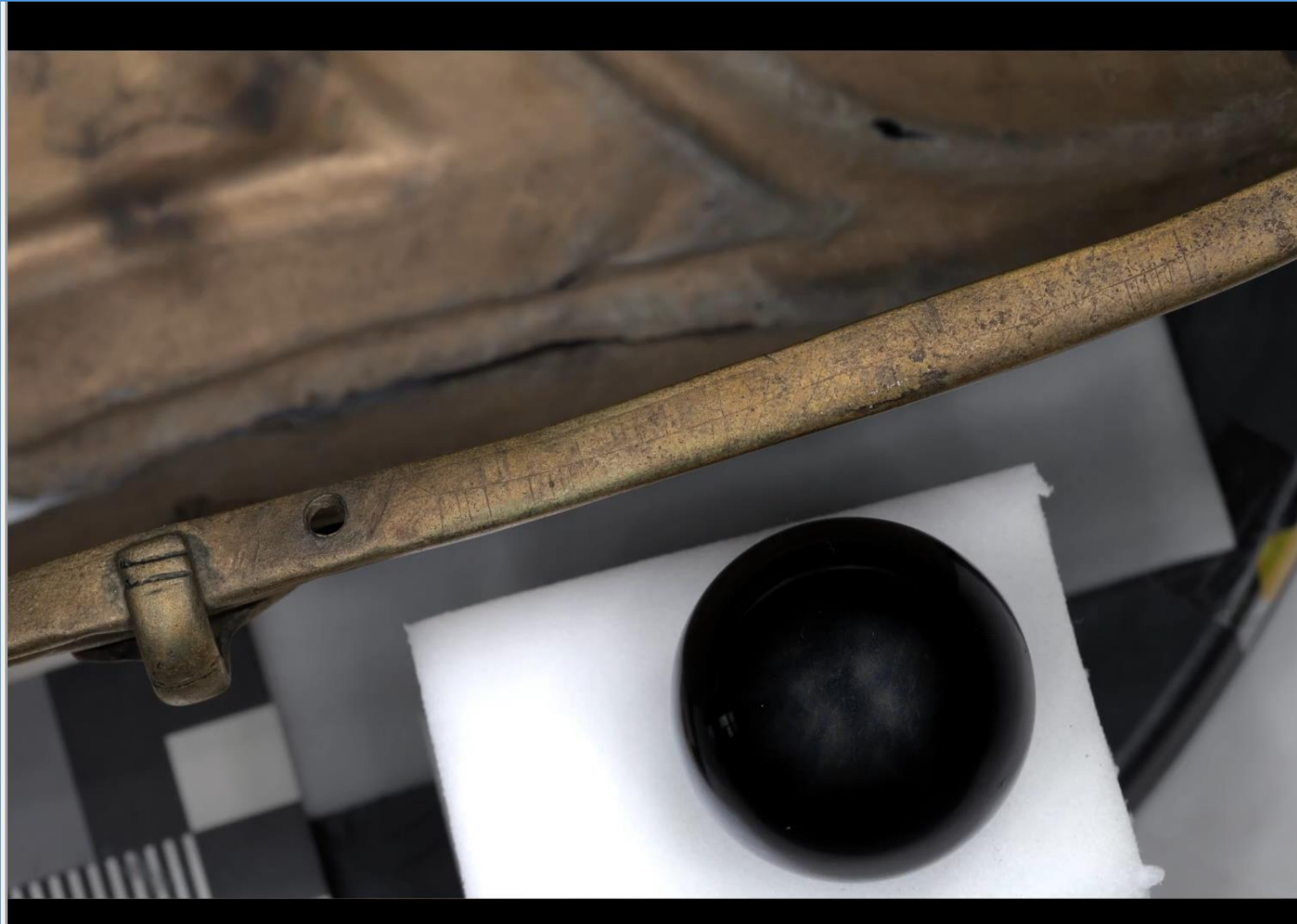


Lighting positions



KU Leuven Bibliotheken

Reflectance Transformation Imaging



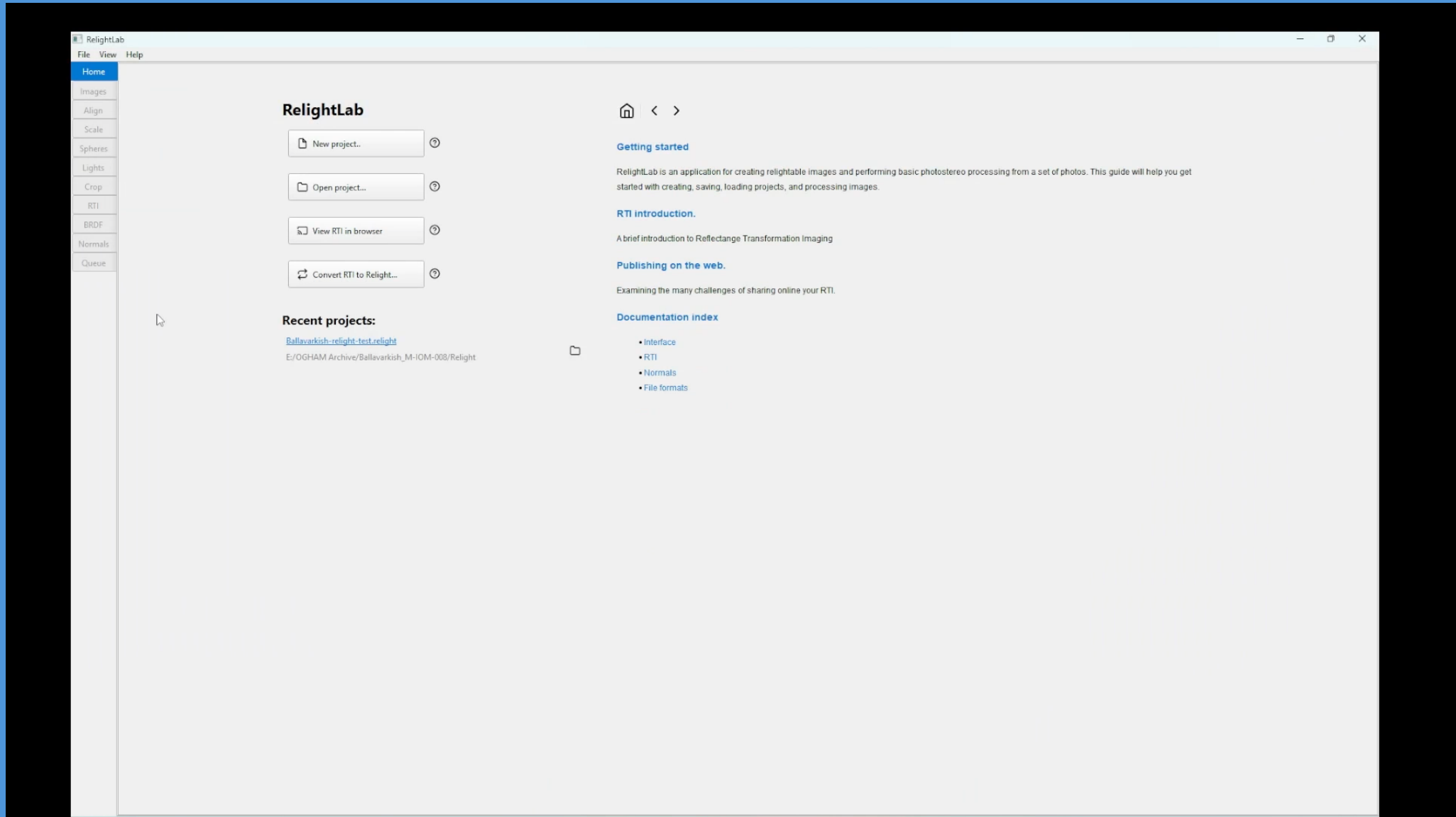
Reflectance Transformation Imaging Setup (HSH for Large Objects)



Unlisted video, link: <https://youtu.be/ekT96S9XF4c>

CHI Imaging: <https://culturalheritageimaging.org/Technologies/RTI/>

RTI: Relight Processing



Unlisted video, link: <https://youtu.be/tFvKkMtLgXk>

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Research Insights from Digital Imaging

Working with outputs

Case studies

Working with Outputs

- Outputs shape what becomes visible, measurable, and discussable
- Digital methods produce representations, not facts:
A photogrammetry mesh, a PSI relighting view, and an RTI file are three different ways of seeing the same object BUT none of them is the object
- Each output emphasises certain features while suppressing others
- Reading digital imaging outputs is a research skill

From Questions to Methods, *and back*

- The research questions and the target material drive method choice
- The methods also reshape questions
- Digital methods often reveal unexpected features, forcing us to reframe the original focus
- Expect iterative research design

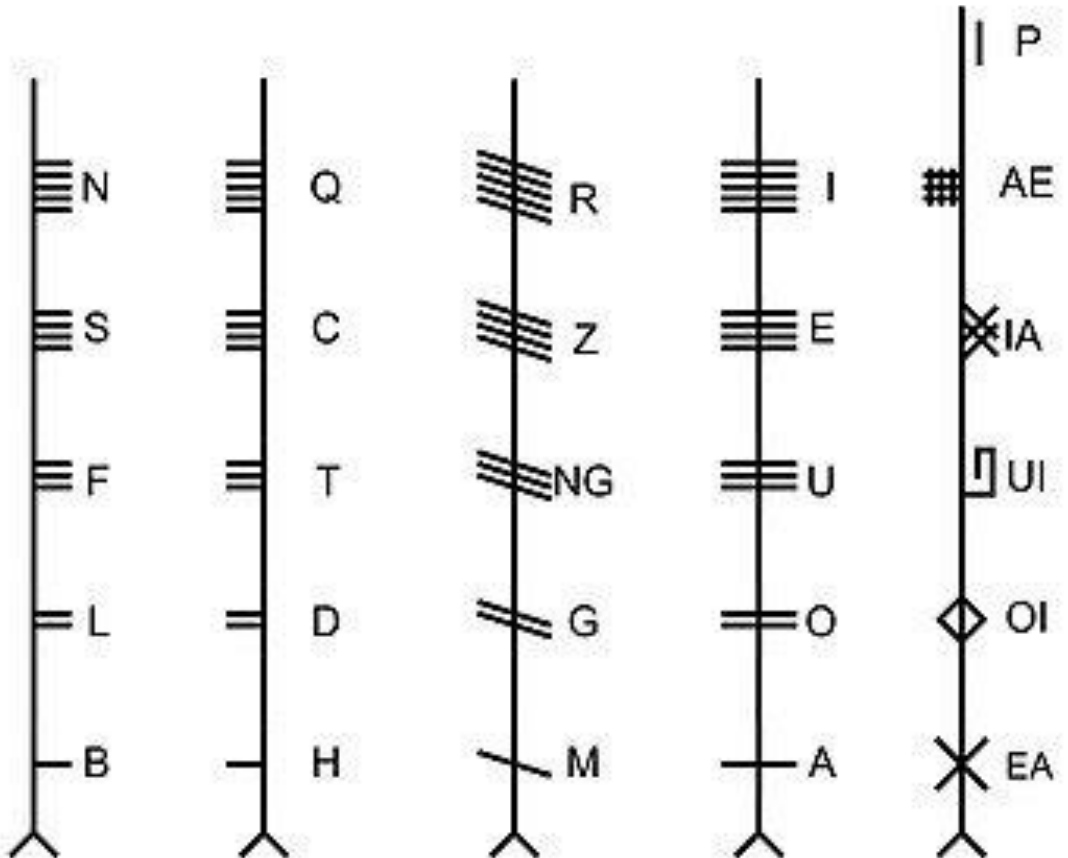


OG(H)AM

Harnessing digital technologies
to transform understanding of ogham
writing, from the 4th century to the 21st

- Script designed for Primitive Irish language, structure based on Latin alphabet

<https://ogham.celt.dias.ie/>



Research Questions

- Comparison of inscriptions
 - Within and beyond modern boundaries
 - Previous records
- Clarification of Inscriptions
 - Ballavarkish (M-IOM-008)
 - Brodie (S-MOR-001)
- Groove Analysis
 - Bi-script stones
 - Newton Stone (S-ABD-001)
 - Colbinstown (I-KID-001)



S-SUT-001 Golspie



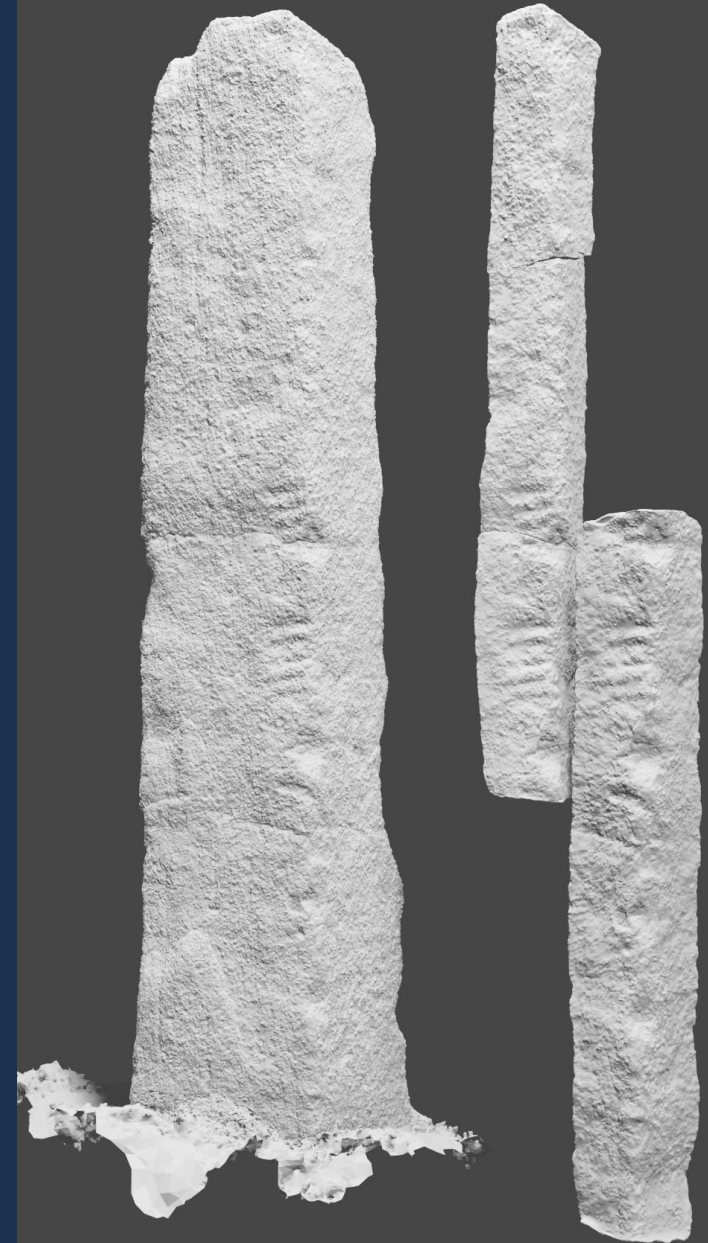
I-DON-001
Magherareagh



Northern Irish ogham stones positioned side-by-side in the same 3D space.
Left to Right: Topped Mountain, Dunalis, Aghascrebagh, Drumconwell, and Gartree

Comparisons with casts

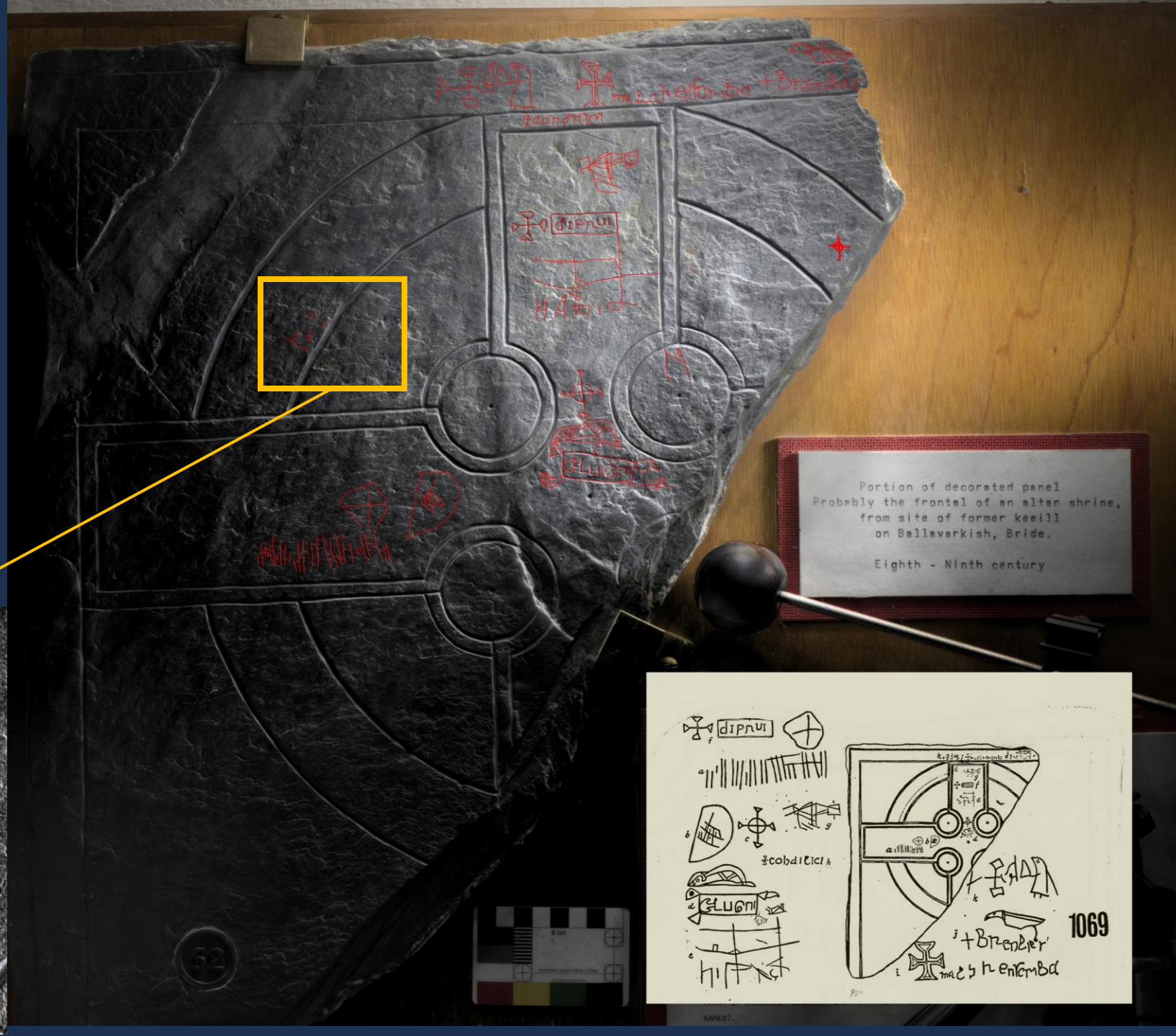
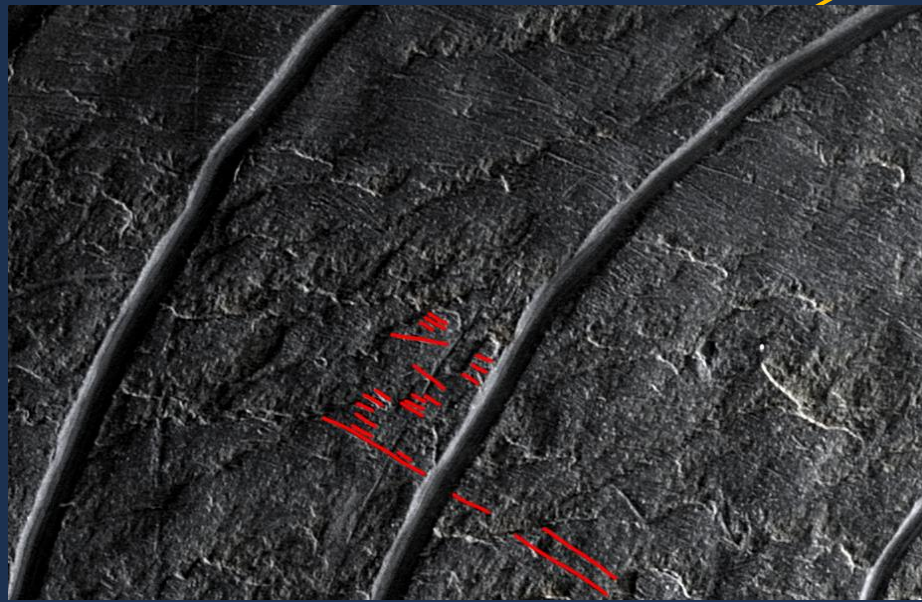
- Gigha
 - HES Archives casts
 - Less weathering/wear since cast was made than expected



S-ARL-001 Gigha vs Plaster
casts in HES Archives

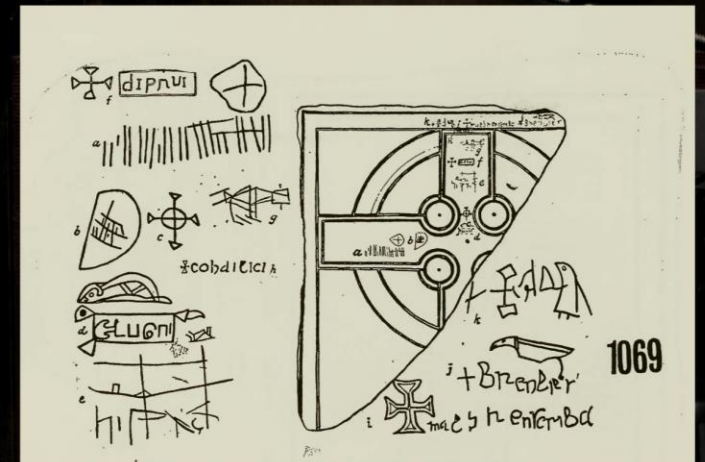
RTI - Ballavarkish

- Graffiti lightly incised – hints of a second previously unidentified ogham inscription

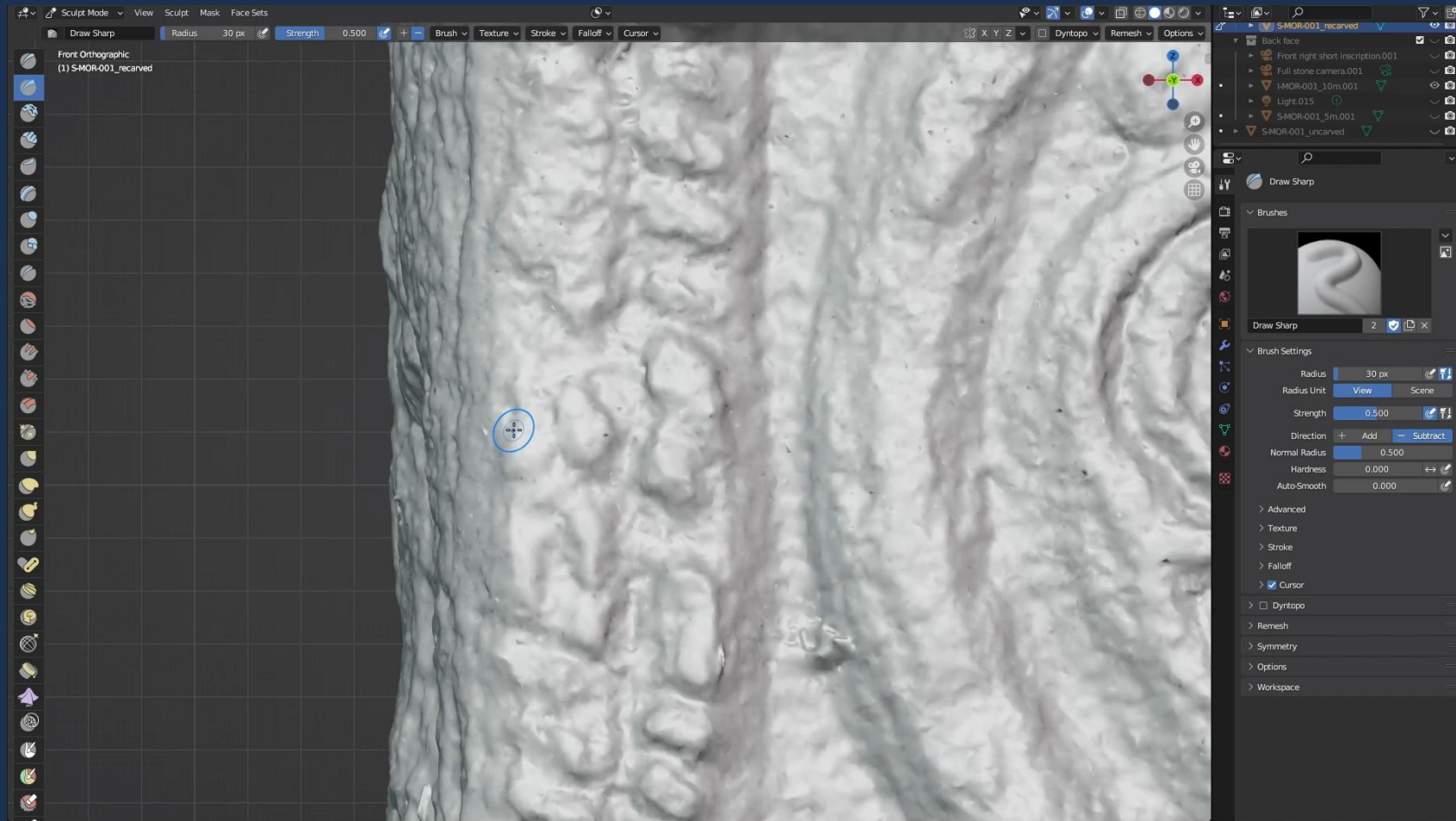


Portion of decorated panel
Probably the frontal of an altar shrine,
from site of former keeill
on Ballavarkish, Bride.

Eighth - Ninth century



Experimental Tools

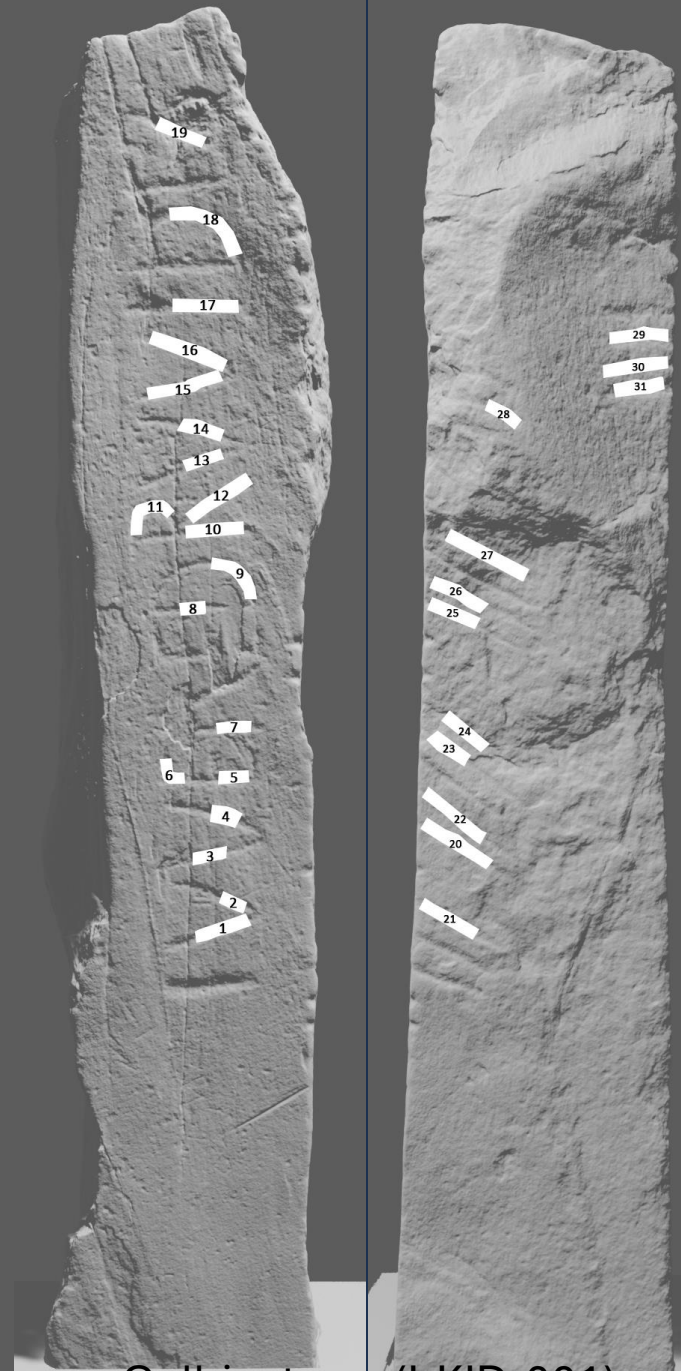


Groove Analysis

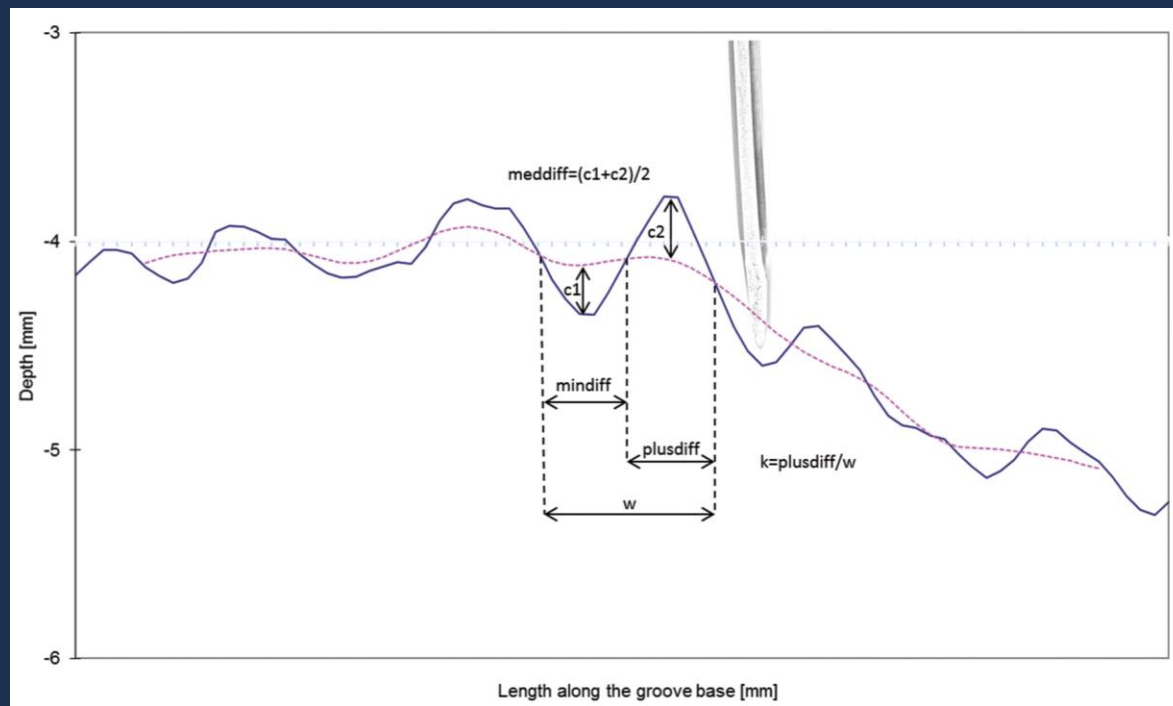
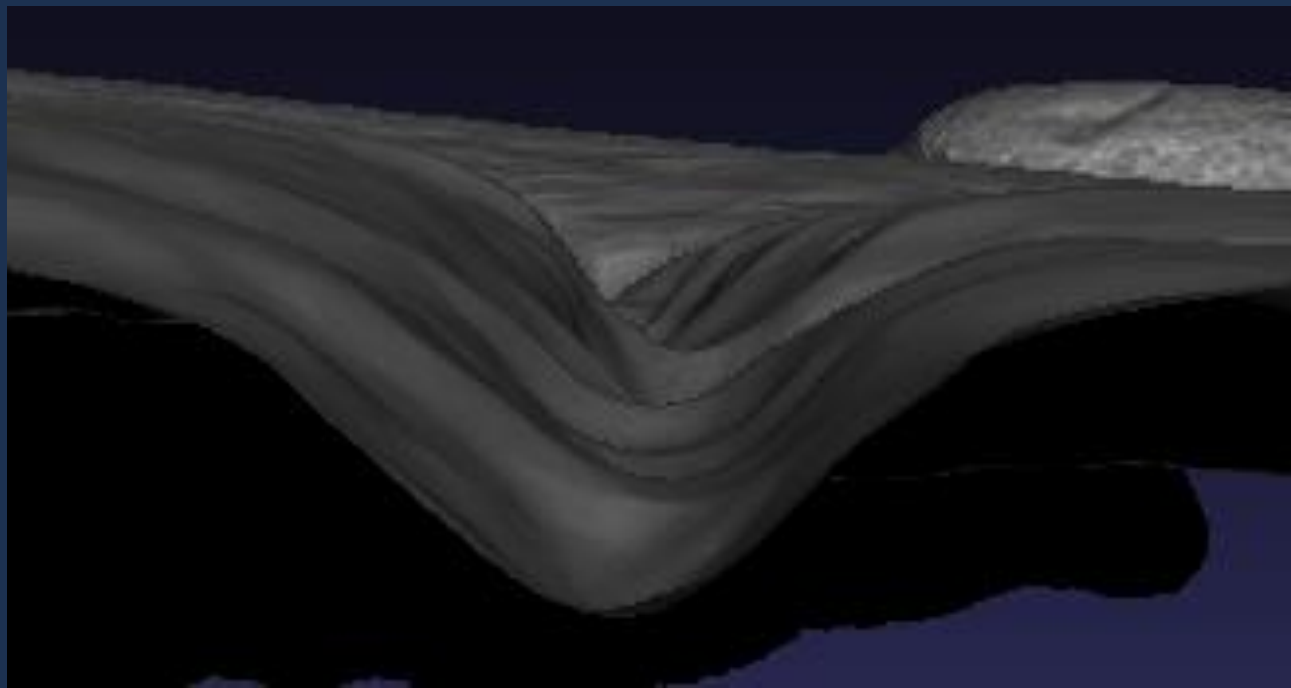
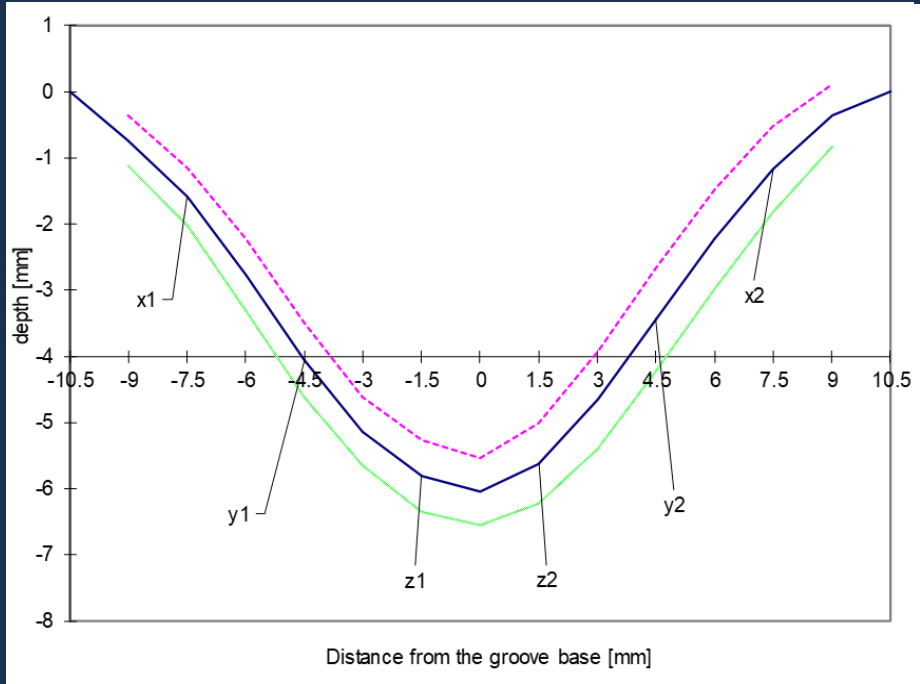
- Bi-script Monuments
- Same carver, or one script first, the other added later?
- Or two contemporary carvers?



Newton Stone (S-ABD-001)

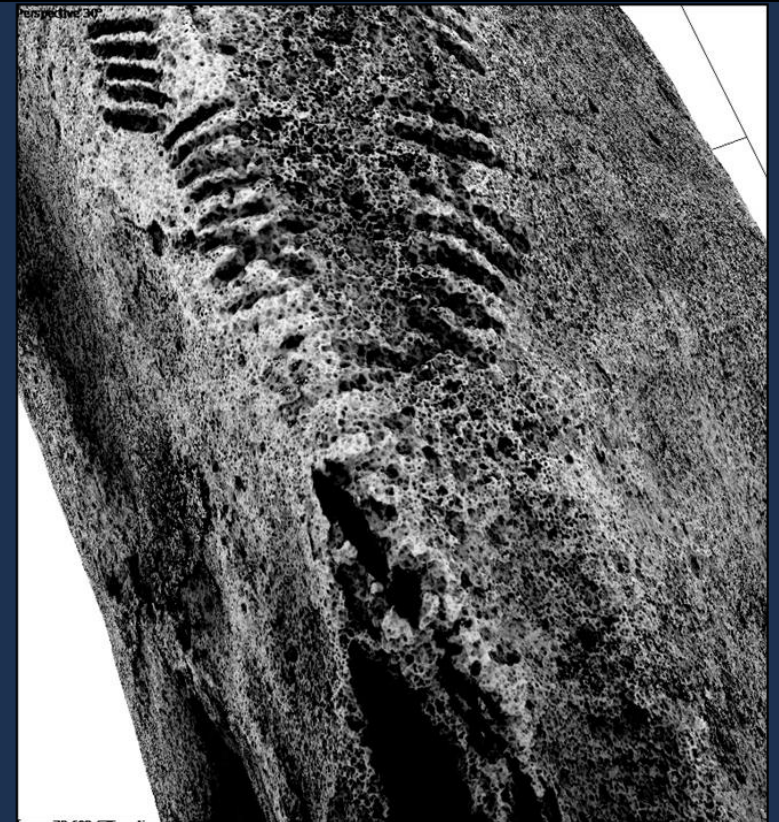
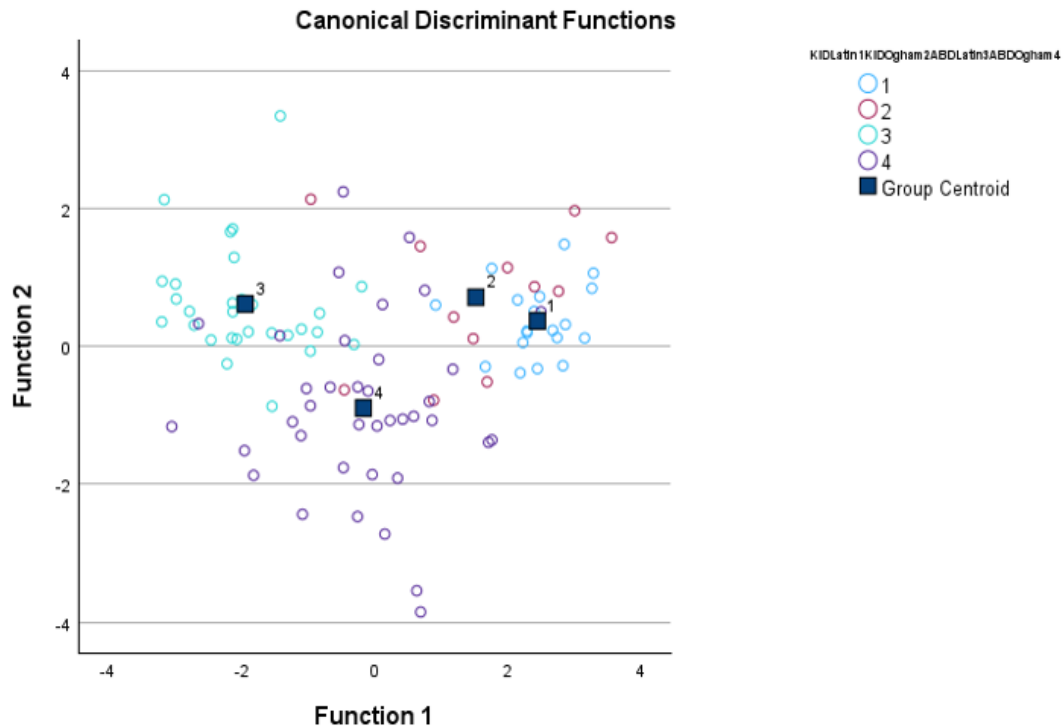
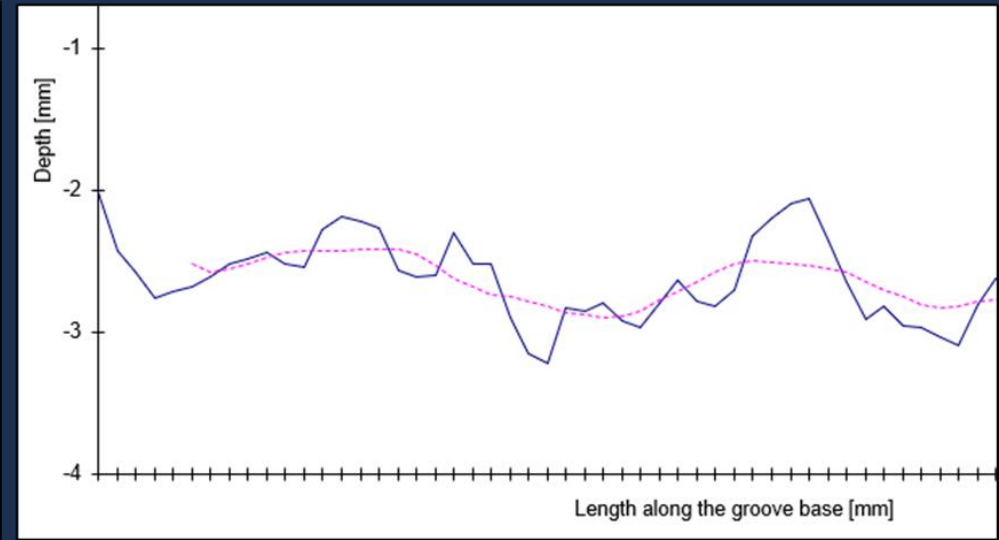
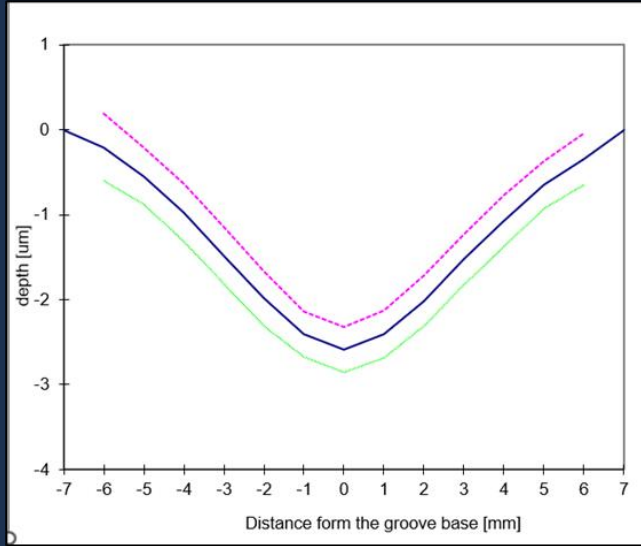


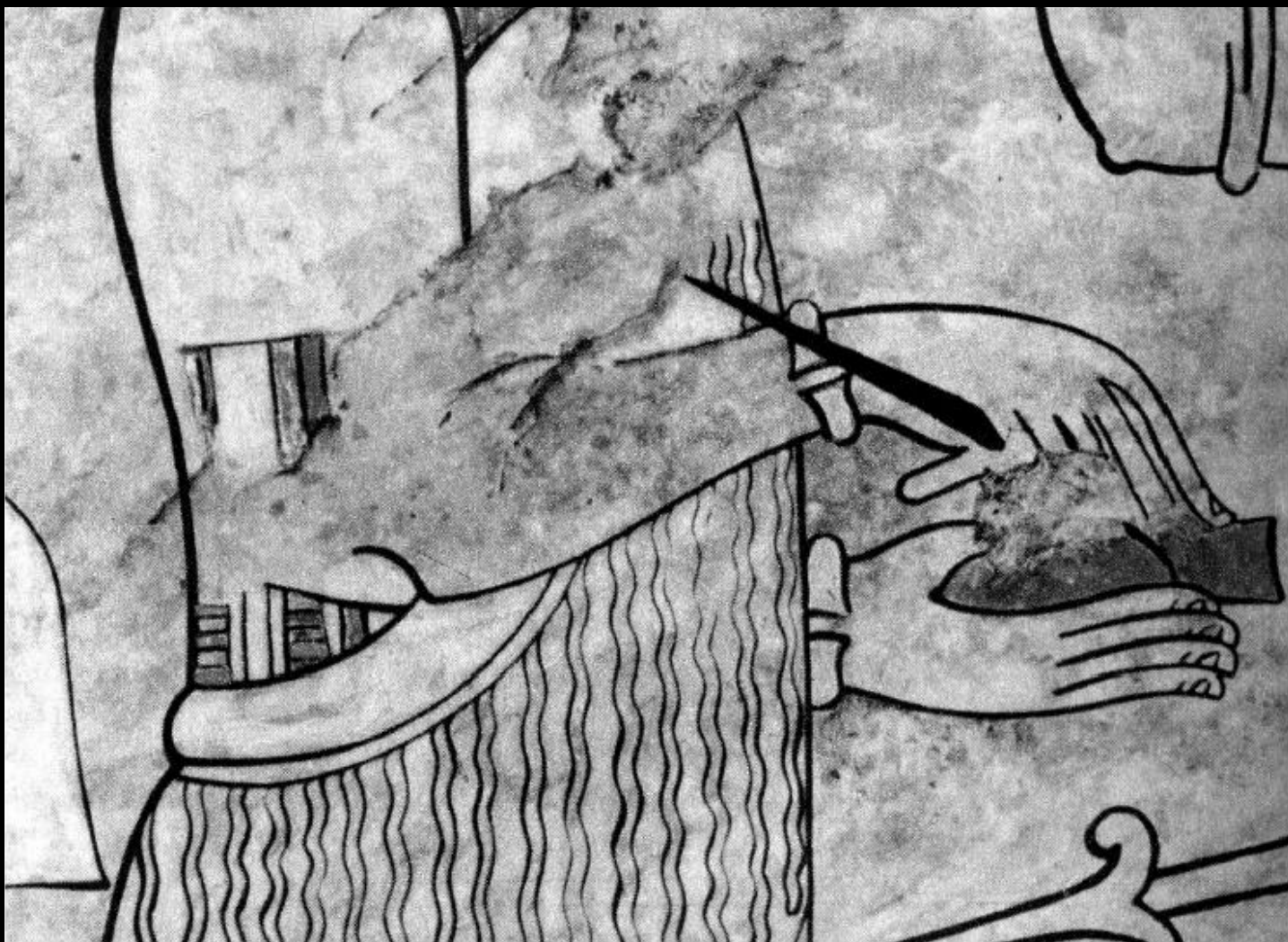
Colbinstown (I-KID-001)



Groove Analysis

- Measurements form basis for ten variables for each sample
- Multivariate Statistics
- Collaboration with Laila Kitzler Åhfeldt



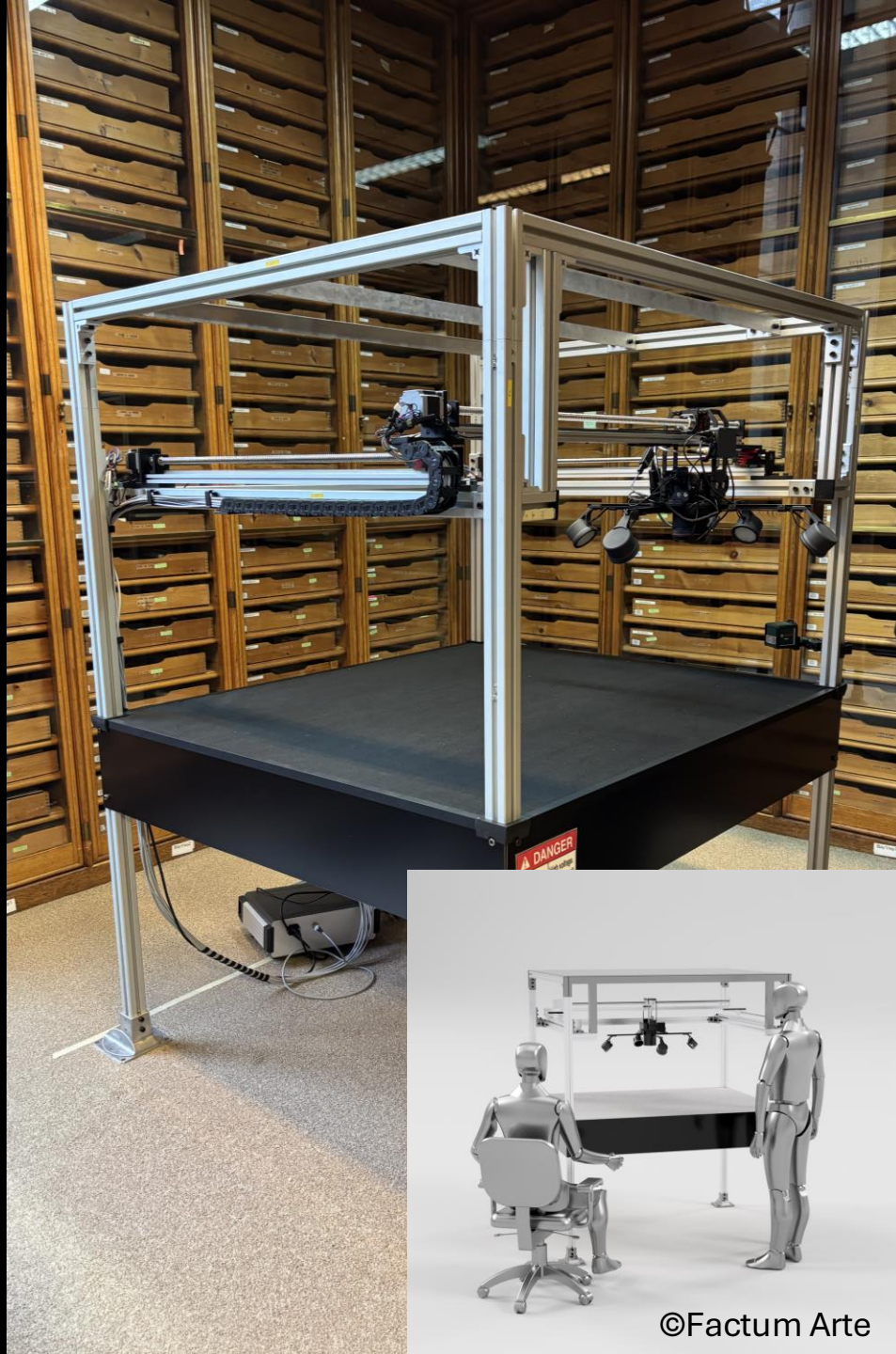


Cuneiform writing system, used in the ancient Middle East c. 3200 BCE to 100 CE

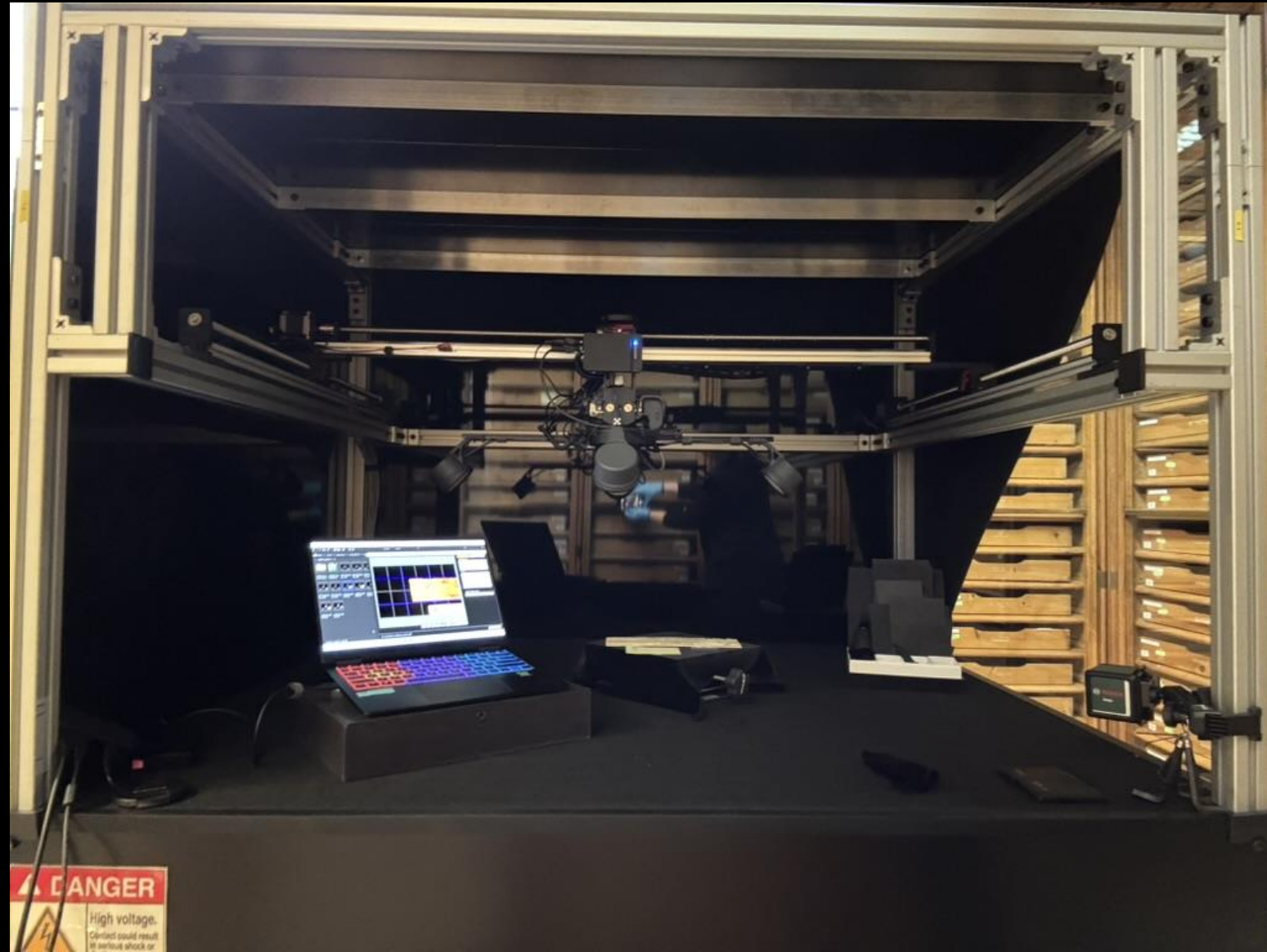
©Alberto Giannese

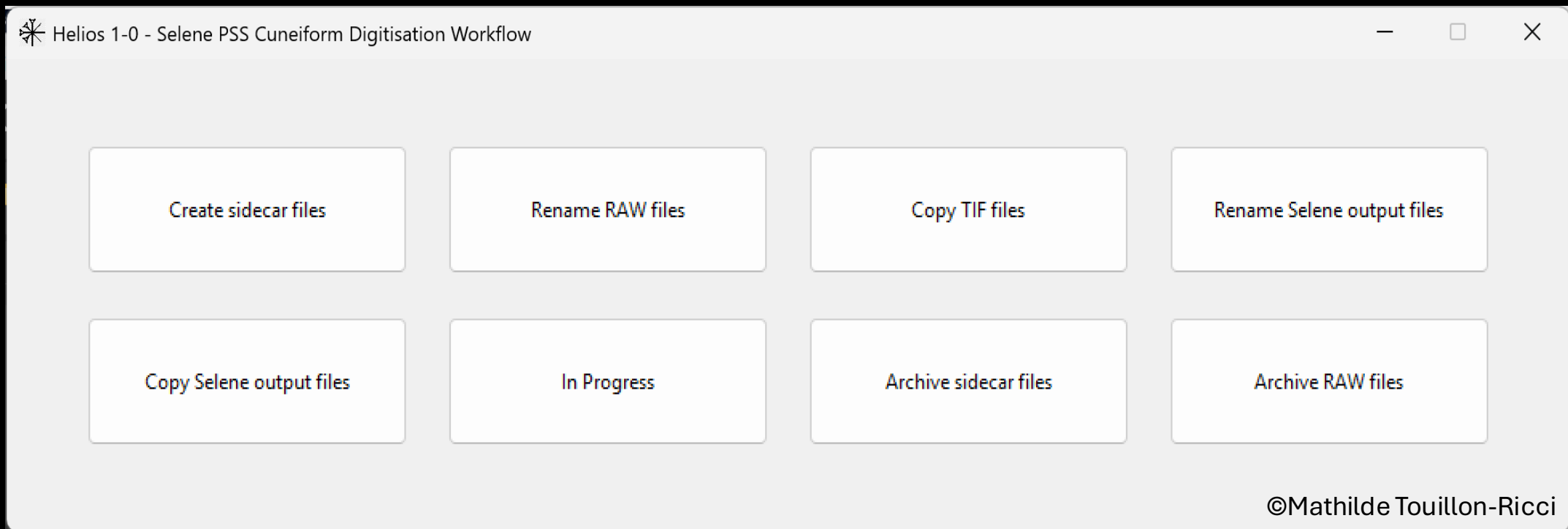
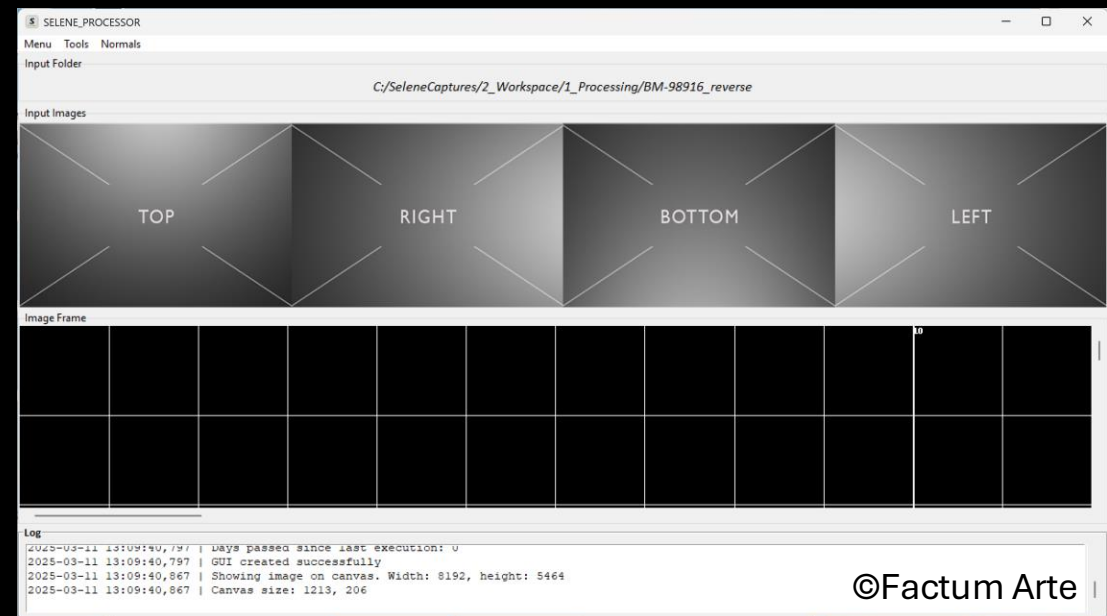
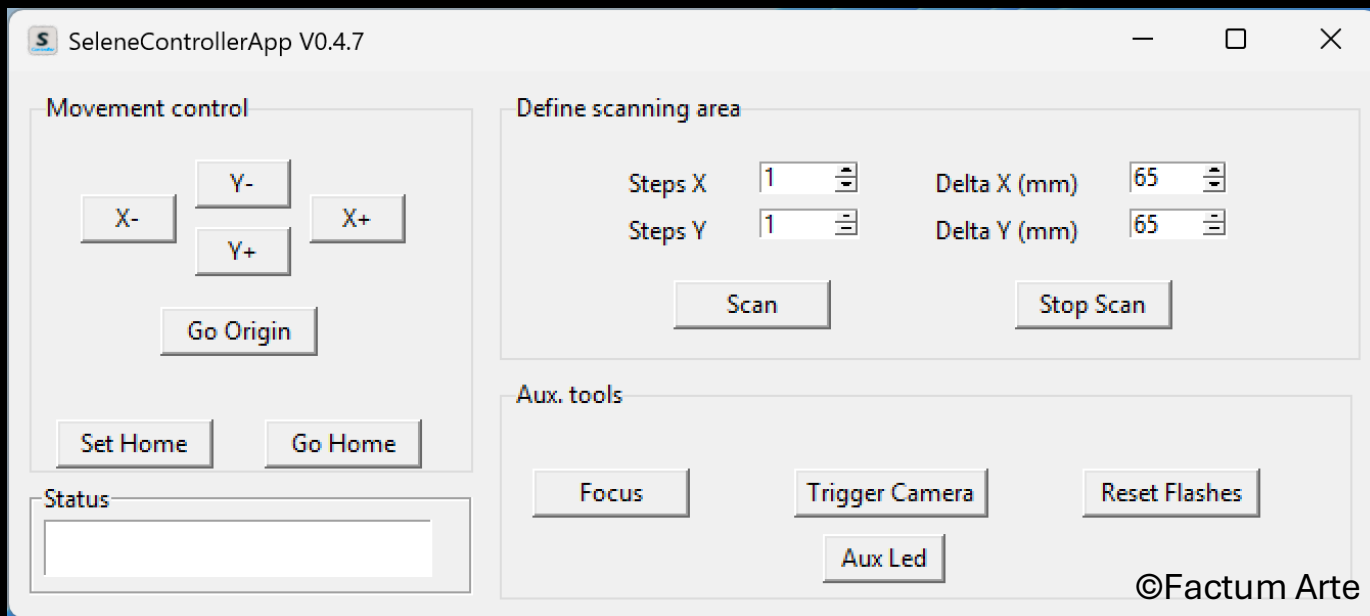
Photometric stereo scanning at the British Museum – Selene PSS

©Mathilde Touillon-Ricci / Trustees of the British Museum



©Factum Arte

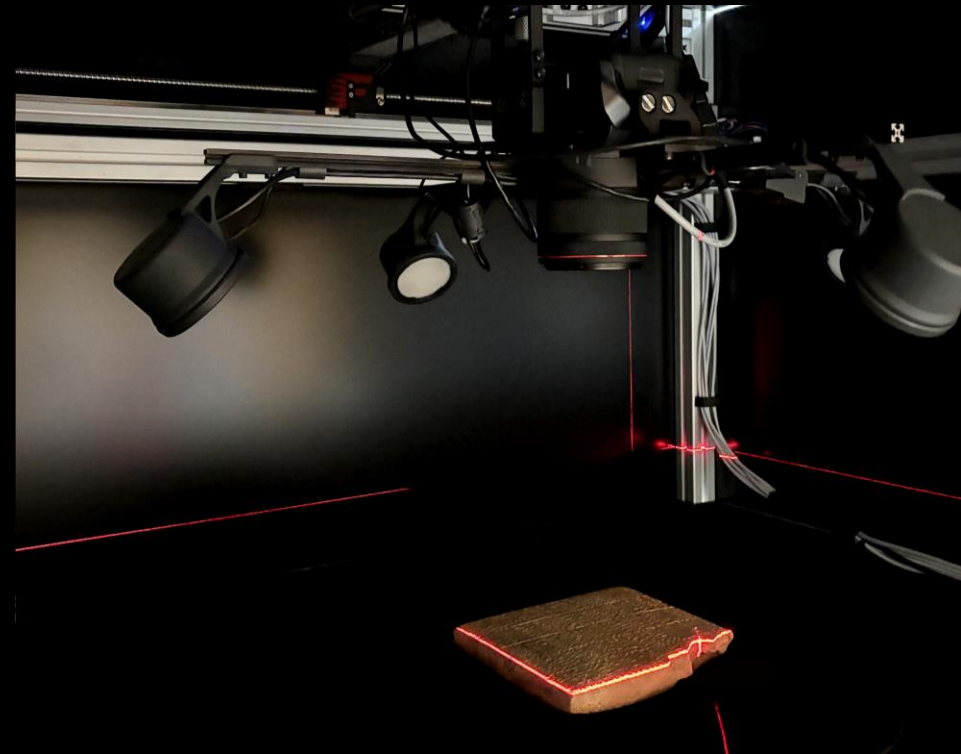




Case Study

Photometric Stereo Imaging meets Ancient History

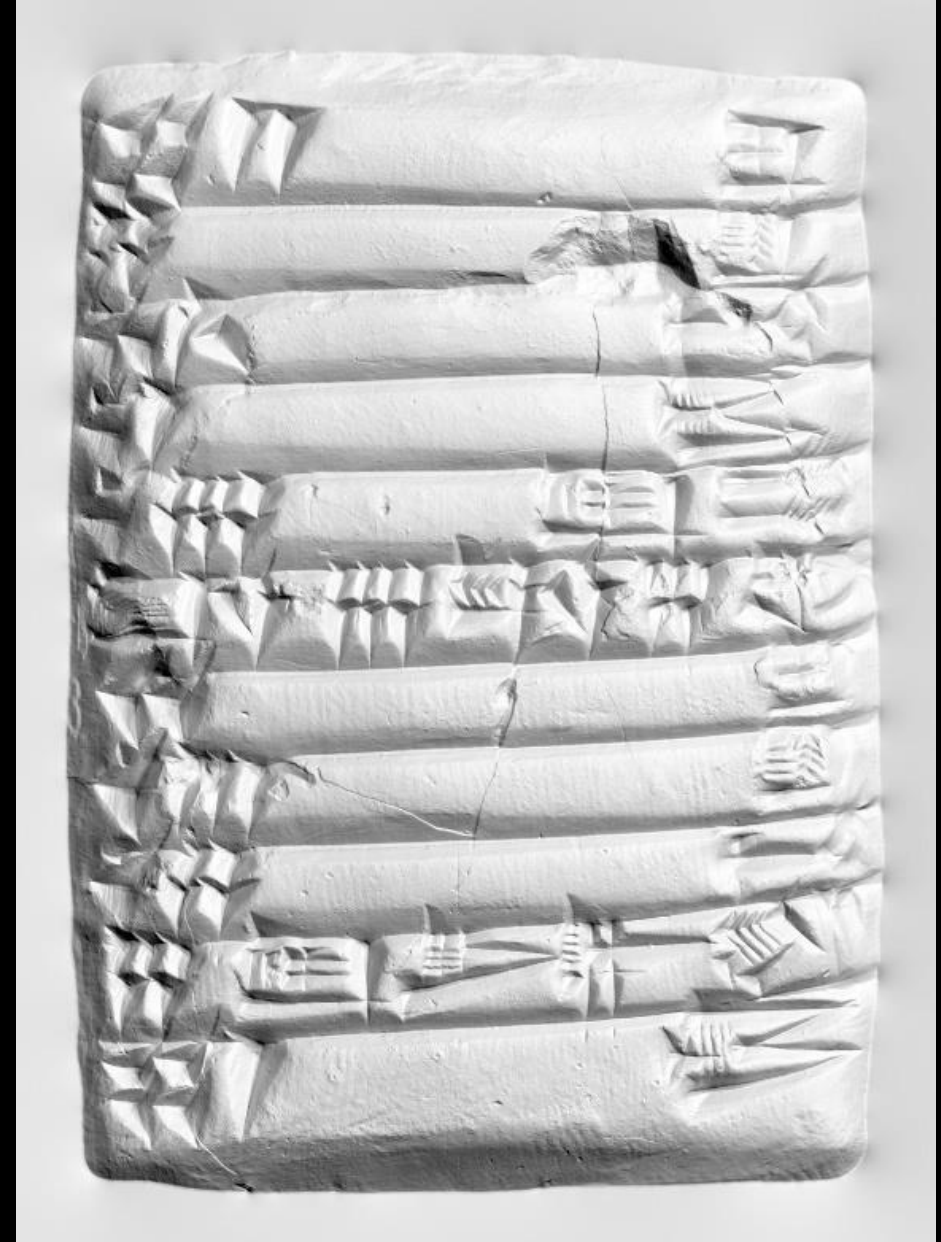
- PSI was tested on the cuneiform collections and used to document surface information that:
 - standard photography does not reliably capture
 - volumetric imaging is not optimised for
- Focus on surface evidence
- Research questions about:
 - Writing process and correction
 - Material traces of production and handling
 - Administrative practice beyond text content



Impressions: Textiles



Stocks record – sheep
removed from the flock –
c. 2050 BCE (BM 12252)
©Trustees of the British Museum



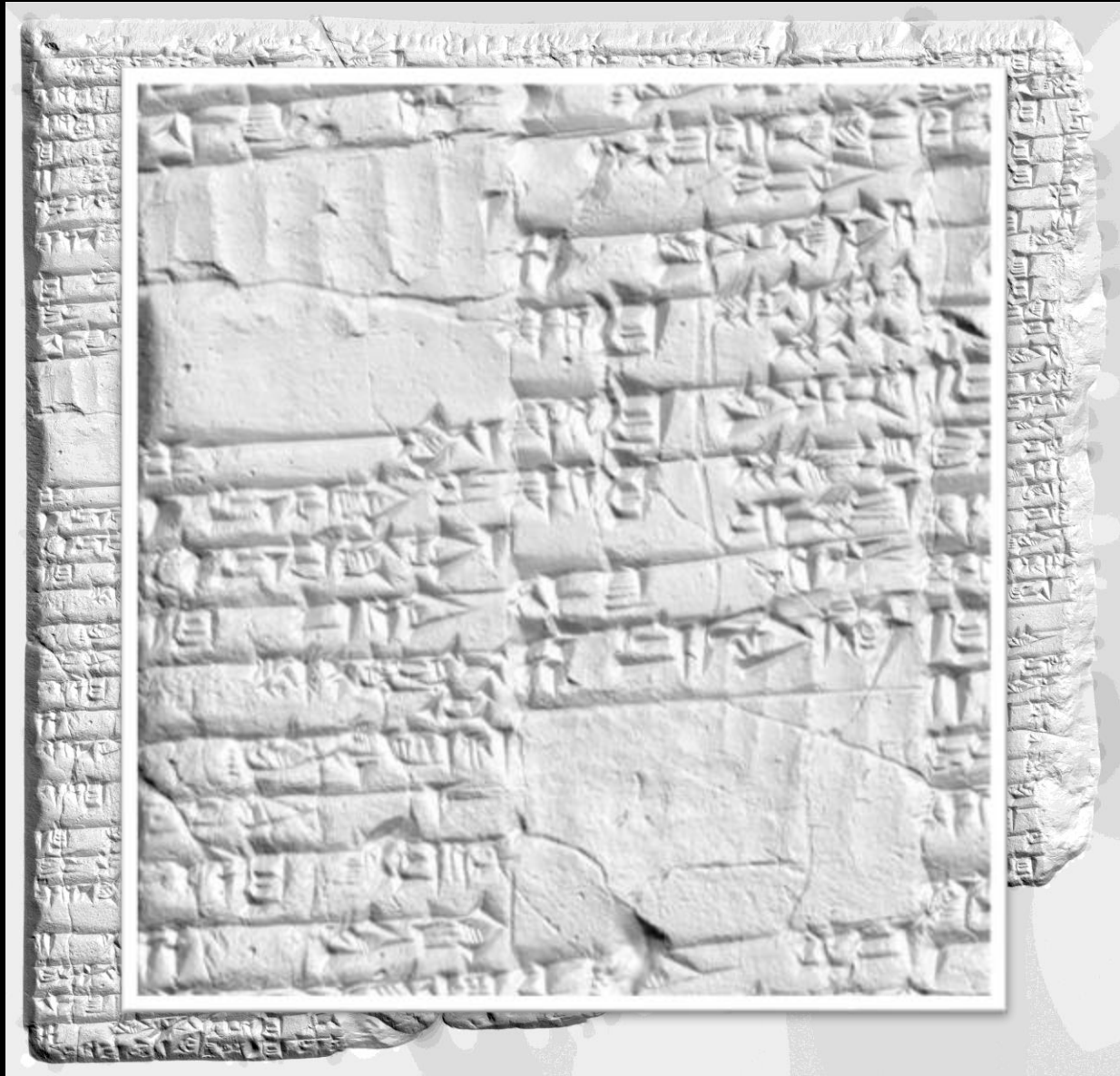
Impressions: Fingerprints



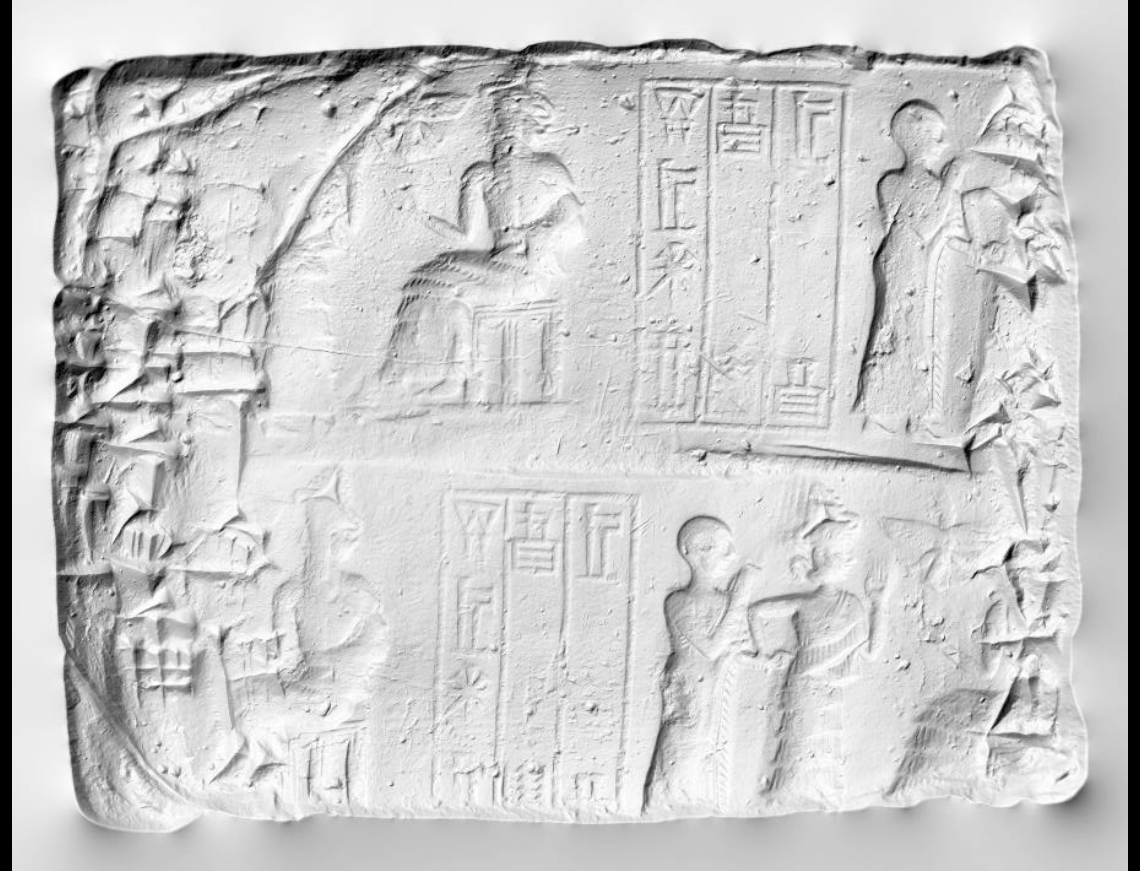
Receipt – c. 2050 BCE (BM 12276) ©Trustees of the British Museum

Erasures





Sealings



Sealed case – c. 2050 BCE (BM 12252 A) ©Trustees of the British Museum



يَا دَابَّةُ امير الوحي وخازن القدس ويقال له ايضاً الروح الامين والروح القدس
والنابغة الامير الوحي وطاهر الملائكة جاء في القرآن ان الله تعالى اذا تكلم بالوحي
سمع اهل السماء صلصلة كجمر المسكوك على الصفا فيسمعون ولا يزالون كذلك
حتى تأتيهم جبرئيل فاذا جاءهم فزع عن قلوبهم قالوا يا جبرئيل ما ذا قال ذلك
فيقول الحق فنادى للوحي وجاه بالخير ان النبي صلى الله عليه وآله وسلم
قال لجبرئيل اني احب ان اراك على صوتك فقال انك لا تطيق ذلك

صلى الله

Manuscript page – The Angel Israfil – c. 1375-1425 CE (BM 1963,0420,0.0.1)

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Welcome to Mirador



Summary and Comparisons

- Photogrammetry and Photometric Stereo

Beyond Imaging

Ethical Considerations

- Cultural and community considerations
- Ownership, copyright, and control
- Platforms and politics of access
- Archiving and data retention
- Environmental responsibility

What do you need multi-dimensional digital imaging for?

Photogrammetry Exercise